

Reimagining development: Experimental evidence of public reception to progressive development narratives in an emerging economy*

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Abstract

We investigate what narratives can influence the public to be more receptive to progressive policies and development ideas centered on sustainability and social justice. We conduct an online experiment in which participants are randomly assigned to a short narrative about the economy. Each narrative appeals to different principles, from equal opportunity to environmental protection. We find that each narrative has its merits and flaws. The narrative that calls for preventing development from displacing people and damaging the environment, along with the narrative that emphasizes human development, shows the most promise in cultivating people's receptivity. Meanwhile, narratives that call for re-balancing economic distribution and promoting equal opportunity have a limited impact and even generate backlash against progressive ideas. We also find that simpler messages are more effective in shifting perceptions. Conjoint experiment affirms that environmental protection is not a paramount concern for the public. Our insights will inform the policy advocacy work of many think tanks and may help civil society organizations wage more effective campaigns.

Keywords: narratives, development, sustainability, social justice, public pressure.

JEL Codes: D83, D63, D91, I00, O10, O44, Q01.

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¹We welcome feedback. Access the latest version [here](#). Online Appendices are available [here](#). Working codes are available in our [GitHub repository](#). Data is available by request via the [NarrativeLab](#) website.

1 Introduction

Throughout history, public pressure has been a key driver that ignites major reforms. Economic reforms are no exception. From a 40-hour 5-day workweek (Davis, 2022) to the right to unionize and collectively bargain (Fleischman, 2021) that has shaped our economy today, public pressure has been a decisive factor in the success of reforms.

Nowadays, the way governments implement economic development in developing countries is an area where public pressure can play a central role. Since the latter half of the 20th century, emerging economies have pursued a development strategy that combines industrialization and natural-resource extraction (Hoskisson et al., 2000). Many of these countries have yet to complete their development. However, it becomes apparent that these countries need a better development strategy. While this conventional strategy has lifted many people out of poverty (Matthews, 2019), it has come at a steep price: ecological crisis (Khan, 2025) and soaring carbon emissions that accelerate global warming (Fuhr, 2021). It has also propelled rising inequality (Lakner and Milanovic, 2016).²

While formulating an alternative development strategy is a technocratic task undertaken by experts, public pressure can ensure that the push toward reform endures. It can make discourse and conversation about reform persist. It can also help sustain reform on policymakers' agendas. Otherwise, policymakers might lack the political will and may neglect reform agendas. There are all the critical conditions for a major reform.

However, the public will only exert pressure for ideals they strive for. One crucial factor influencing people's ideals is the narratives to which they're exposed (Beach and Wise, 2022). Studies have shown that narratives can influence public discourse, trigger emotions, shape public opinions, beliefs, attitudes, behaviors, and, in turn, affect policy formulation (Smyth, 2024; Algan et al., 2025; Stantcheva, 2024).³ As such, we must find the right narrative that can motivate public pressure.

In this study, we aim to identify the narratives that can arouse public support for progressive development ideals and policies centered on sustainability and social justice. We run a randomized online experiment exposing participants to four different narratives about the economy. Each narrative emphasizes distinct values or ideals that the new development approach should embody, from social equality to environmental preservation. We conduct the

²Furthermore, the advent of novel technologies, such as AI, will substantially reconfigure the landscape of work and income distribution in the near future (Acemoglu, 2021; World Bank, 2019).

³The conventional development strategy became the dominant one because the narrative about it was also dominant at the time, as it was the policy instrument prescribed by many leading economic advisory institutions (Garner, 2004; Klein, 2004; Haddad, 2008; Haddad et al., 2011; Cherif et al., 2024). Another example: Americans who believe that the "American Dream" narrative still holds are more likely to believe that upward social mobility is more achievable than reality suggests and less likely to support redistribution (Alesina et al., 2018, 2020).

study in Indonesia, one of the largest middle-income nations that is fighting two battles simultaneously: raising the living standards of 176 million people⁴ while limiting its carbon footprint and mitigating ecological disasters ([Hapsari et al., 2024](#)).

We find that each narrative has its strengths and drawbacks: it may stimulate desired changes in beliefs and attitudes but also generate unintended backlash against progressive ideas. We highlight three key findings. First, the “No victimization” narrative elevates support for inclusive, rights-based development priorities and heightened sensitivity to the human costs of displacement, without eliciting any detectable backlash. Second, the “Balanced development” narrative increases the salience of education and basic services but inadvertently undermines support for infrastructure-related and some clean energy policies. Third, narratives that call for reforming redistribution (“Fix the distribution”) or accentuate stark differences in privilege (“Equal opportunity”) have limited positive effects and, in some cases, provoke counterproductive responses.

We then trace the possible sources of the backlash to each paragraph in the narrative script and find that cautious wording is necessary to minimize the risk of misinterpretation. Relatedly, simpler, more focused messages foster better comprehension, while longer, more complex narratives risk dilution and misunderstanding. The conjoint experiment confirms that the equal (rather than equitable) distribution of income gains remains the paramount development outcome for Indonesians. In contrast, the environmental impact of development does not appear to be a major concern for the Indonesian public.

Our findings hold practical implications. Think tanks can utilize our insights to shape their policy advocacy strategy. Civil society organizations (CSOs) can learn from our insights to design and wage more effective public campaigns to encourage citizens to participate in collective action. All these efforts are crucial to ensure the appetite for reform persevere.

As we show in this paper, we must be judicious in harnessing the power of narratives, as their framing can determine the direction of public pressure. Yet, despite the potent force of narratives, economics still lags behind other disciplines in its study ([Shiller, 2017](#)).⁵ While studies on the impact of specific policy narratives are on the rise ([Stantcheva, 2020](#)), we still observe scarcity in the investigation of the effects of broader economic development narratives ([Roos and Reccius, 2024](#)); in particular, how narratives around the principles and values that govern the economic development model influence one’s conception of economic governance, i.e., how the economy should work ([NEON et al., 2018](#)). Furthermore, while narrative study is a nascent and growing field in economics ([Haaland et al., 2025](#)), controlled experiments examining the impact of narratives such as ours remain rare ([Haaland et al., 2023](#); [Shiller, 2017](#)),

⁴Considered as “poor” using the World Bank’s upper middle income country’s poverty line ([World Bank, 2025](#)).

⁵See the seminal work of [Shiller \(2017\)](#) for some reviews of narrative studies outside of economics.

especially in a developing country context.⁶

To date, we are aware of only three other related studies in Indonesia: [Sim et al. \(2024\)](#), [Marshall et al. \(2023\)](#), and [Dechezleprêtre et al. \(2025\)](#). The first examined messaging to persuade the public to support efforts to prevent land subsidence. Yet the researchers did not investigate the impact of different messages, but rather the effects of different messengers and the focus of information of the same message. The second and third studies evaluated the impact of various climate narratives. Our research builds on this study by addressing the narrative more comprehensively and broadening the message’s focus to encompass both environmental protection and social justice. This may have broader implications for people’s fundamental understanding of how the economy and development should work ([NEON et al., 2018](#)).

Our study also contributes to the broader discussion on messaging interventions to alter people’s beliefs and behaviors. The most commonly studied type of messaging in economics research is information intervention ([Haaland et al., 2023](#)). However, learnings from past studies and efforts to shift people’s perceptions suggest that information alone may not be sufficient, as interventions that directly attempt to educate or correct people’s misperceptions or misinformation tend to fail ([Singal, 2014](#)).⁷

We find an indication that information and narratives are complementary. Overlooking the narratives that shape the cognitive lens through which a person interprets information can dampen efforts to change beliefs and attitudes ([Beach and Wise, 2022](#); [Stantcheva, 2024](#)). Narrative provides a frame for the information and helps the public interpret it as reformers desire. Yet, without accounting for the knowledge and information an individual currently possesses, narratives can backfire, misdirecting people into counterproductive views.

We organize the remainder of the paper as follows. Section 2 further elaborates on the study context. Section 3 describes the survey design, while Section 4 complements it with our empirical framework. We then present the results in Section 5. We discuss the implications of the findings in Section 6, and Section 7 concludes the paper.

⁶For the developed country context, the closest related work to ours is by [NEON et al. \(2018\)](#), who, similar to us, examined how different economic narratives influence people’s understanding of the economy in the UK. We also highlight the notable work of [Stantcheva \(2021\)](#) and [Stantcheva \(2022\)](#) on policy narratives about taxation and trade in the US, respectively. Taken together, the studies found that varying the narrative emphasis of key policy aspects can lead to different policy views ([Stantcheva, 2020](#)). This exemplifies the importance of narratives in driving policy discourse.

⁷[Hoy et al. \(2024\)](#) found that informing Indonesians of the true level of inequality in the country raises concerns about inequality but does not sufficiently rally public support for redistribution policies. The same study also found that informing Indonesians of their relative position in the national income distribution reduces support for redistribution. Similar challenges arise in environmental policy. Even after learning the correct definition of climate change, 60% of Indonesians remain willing to permit deforestation for essential economic projects ([Leiserowitz et al., 2023](#)). Nonetheless, [Evans \(2019\)](#) shows that there is some evidence to the contrary.

2 Study context

While novel suggestions to alternative economic development approaches have emerged, they still lack widespread adoption ([Gronchi et al., 2021](#)). As with any episode of creative destruction, history has taught us that new economic models that disrupt established ones will attract opposition ([Acemoglu and Robinson, 2012](#)). Economic transformations often involve inconvenient but necessary changes to how people live their lives ([UNDP, 2017](#)).⁸ While we must assist those who lose out from the reform to make their transition too ([UNDP, 2021](#)), this change can nonetheless be uncomfortable, and people might perceive this as an invasion of their way of life and beliefs ([Anderson, 2024](#); [Tasch, 2024](#)).⁹ This can make people reluctant, and even resistant, to reform agendas ([Harring, 2021](#)). Policymakers, in turn, might hesitate to introduce reforms when they do not see electoral incentives.

Indonesia clearly showcases this problem. Our earlier study shows that (and this study confirms it), unless they have experienced environmental degradation and social injustice due to development firsthand, environmental protection and social equality are treated as afterthoughts when most people imagine what constitutes successful development ([Bahana and C4C, 2025](#)).¹⁰ It is not surprising then that the majority of Indonesians are not yet engaged in tangible activism ([Leiserowitz et al., 2023](#)). In addition, worse than policymakers who lack electoral incentives to reform, our earlier study found that the government has actually played an outsized role in shaping public narratives and opinions about development ([Bahana and C4C, 2025](#)).

The situation above complicates efforts to build public pressure for progressive reforms. Instead of collective citizen action, we have public omission carefully nurtured by the government, leaving harmful development practices relatively unchallenged. As such, we must more effectively persuade the public to achieve buy-in for progressive development goals and reform ideas.

This is where the urgency for an empirically tested new economic narrative that centers on sustainability and social justice agendas arises. Narratives can provide the right framing to make unpopular policy ideas or alternative models seem attractive ([Marshall et al., 2023](#)). When disseminated through public campaigns, consistent narrative-building efforts can con-

⁸In the energy transition context, for example, a successful phase-out of fossil fuels and non-renewables will eventually lead to workers in these industries being laid off or moving to new jobs ([Parks and Baran, 2023](#)).

⁹Present bias can further complicate this transition, as people tend to overvalue the immediate costs of change while undervaluing the long-term benefits ([Laibson, 1994](#)).

¹⁰In line with this, [Leiserowitz et al. \(2023\)](#) found that 60% of Indonesians agree that forest protection should be deprioritized in favor of projects such as infrastructure development. [Ipsos \(2025\)](#) found that a sizable portion of Indonesians believe that limiting fossil fuels could harm the economy more than climate change itself, and that rapid adoption of renewable energy might cause power outages. [Hoy and Mager \(2021\)](#) found that redistribution is not a paramount concern for many Indonesians.

vert uninformed or indifferent citizens into concerned citizens, concerned citizens into standby citizens, and standby citizens into active citizens ([Amnå, 2010](#); [Amnå and Ekman, 2014](#)). This will lead to the buildup of a stronger civil society capable of exerting public pressure on critical reform agendas ([Centola et al., 2018](#)).

3 Study design

3.1 Sampling and recruitment

We recruited 3,638 research subjects through an online survey partner, [Dynata](#), from late February 2025 to early March 2025. Dynata has a respondent panel database registered on its platform. They send online survey participation invitations to potential respondents' devices, and panelists can decide whether to participate. Panelists complete the survey on their devices in exchange for Dynata reward points.¹¹

We recognize that online surveys are subject to inherent selection bias due to technological limitations. As such, we implement a sampling quota based on sex, age, and province in recruitment to bring our sample closer to national representation. We present key sample characteristics and compare them with those of the population in [Figure A.1](#).¹²

The figure aptly demonstrates that our respondents can be summarized as urban, productive age Indonesians with at least a high school degree. The figure indicates that the sample closely aligns with population benchmarks for gender and regional demographics for which quotas are explicitly applied. However, notable deviations emerge across other characteristics, most notably in age, education, and residence.¹³

These biases, particularly in education and urban residency, are common in Indonesian online surveys and are likely driven by unequal internet access, which disproportionately excludes rural and lower-educated populations ([Stantcheva, 2023](#); [Wati et al., 2024](#); [Yasmine et al., 2025](#)). Accordingly, we advise caution in generalizing the results to programs and interventions.

Nevertheless, we believe that this sample remains relevant for program applications. The

¹¹Median survey completion time is around 19 minutes. To protect data quality from survey speeders, our survey partner automatically excluded survey completions under 12 minutes. Our final sample consists of individuals who passed this screening.

¹²We use Indonesia's National Socioeconomic Survey (Susenas) from March 2023 to measure the population characteristics. Susenas is a biannual, nationally representative, in-person survey administered by Statistics Indonesia, and Indonesia's leading source for tracking socioeconomic indicators.

¹³Older individuals (aged 55 and above) are underrepresented, while middle-aged respondents (25–54) are overrepresented. Compared with their national shares, the sample also overrepresents higher-education graduates by more than 40 percentage points and urban residents by almost 40 percentage points. Household size in our sample is somewhat similar to that of the population.

middle class is often considered a politically influential group. They tend to have higher political interest and engagement than other income groups, frequently discussing politics and being more proactive in expressing aspirations (Wihardja and Wibisana, 2024). In addition, the middle class, particularly in urban areas, tends to have greater access to resources and information, positioning it as low-hanging fruit and a critical actor in the dissemination and uptake of development initiatives (Chun et al., 2017). Their relatively higher digital literacy and internet connectivity make them more likely to engage with online content, participate in discussions, and amplify campaign messages (Wihardja and Wibisana, 2024). As such, their significant presence in public discourse enables them to exert political influence.

3.2 Experiment design

3.2.1 Main experiment

We ask Dynata to randomly assign participants to a control group and four treatment groups.¹⁴ Table 1 presents the group size of each experimental arm.¹⁵ Each treatment group receives a different stimulus in the form of a short Indonesian text narrative.¹⁶ We develop four narratives, drawing on the dominant development narratives identified in an earlier study (Bahana and C4C, 2025).¹⁷ Table 1 summarizes the mapping of the dominant and counter-narrative that we test in this study, as well as the full text of the English translation of the stimuli. The “Dominant narrative” column identifies the existing narrative that the new narrative intends to challenge.

The “Fix the distribution” narrative addresses the government’s blatant dismissal in development discourse that everyone is assured of eventually enjoying the economic benefits somehow. This line of thought is akin to the infamous yet historically mainstream trickle-down economics narrative among Indonesia’s development planners (Hainsworth, 1980; McGill, 2018).

¹⁴As committed in our [AEA Trial Registry](#), we have another narrative group in the survey. The additional narrative is not about development but about the role of the state and the relationship between the citizen and the state. Given the substantial theme differences between this narrative and the other four narratives, and considering that this narrative focuses on a distinct set of outcomes not covered here, we present the analysis of this narrative in a separate paper on democracy and civic space in Indonesia.

¹⁵We calculate power prior to data collection to determine the appropriate sample size. Please refer to our pre-analysis plan at [AEA Trial Registry](#) for the detailed discussions.

¹⁶Participants may read the stimulus as long as needed and may return to the stimulus page at any point during the survey. We pre-tested these stimuli in several FGDs to improve comprehensibility.

¹⁷In this earlier study, we mapped the dominant development narratives in Indonesia through a quantitative content analysis of government officials’ speeches from the New Order until the Jokowi era. We also conducted FGDs with laypeople in several locations with diverse development experiences to understand how they conceptualize “development”. We then used the results of the FGDs and the quantitative content analysis to facilitate a Delphi discussion with academics and practitioners from diverse backgrounds to formulate the building blocks of the alternative narrative.

Table 1: Treatment assignment and sample size

Treatment	Dominant narrative	New narrative	Complete text of the stimulus (translated)	Sample size
1	Economic benefits of development will eventually spill over to everyone	Fix the distribution: Fix the distribution of development benefits	What is the point of Indonesia developing if only a few people benefit from it? For too long, the benefits of development haven't reached everyone. The rich and powerful continue to enjoy the benefits, while the less fortunate remain impoverished. The government needs to do a better job of sharing the benefits of development. They must ensure that all Indonesians can benefit from these advantages. If the government can share development benefits more equitably, then everyone in society, regardless of their location, can achieve a better life.	681
2	Development requires sacrifice	No victimization: Development should not victimize the people and the environment	What's the real point of Indonesia's development when so many people become victims because of it? For the sake of new projects, forests are cut down. For the sake of new projects, many people lose their homes or their livelihoods. This isn't actually progress; it's going backwards. We might get good jobs in factories and mines built on cleared land. But what's the use if we get sick from pollution? How will our children live in the future if there are constant floods or landslides? And if someone else is forced out of their home today, what makes us think it won't happen to us, too? The government needs to make sure that no one's life or the environment is harmed for the sake of development.	702

Treatment	Dominant narrative	New narrative	Complete text of the stimulus (translated)	Sample size
3	Development means building large infrastructure	Balanced development: Development should be balanced in its focus on infrastructure and human development	What's the use of having grand bridges and massive buildings like those in rich countries if our people aren't as smart or skilled? People become smarter and more skilled when their education improves. School graduates will be able to solve problems, generate new ideas, be disciplined, and work effectively with others. When people are well-educated, they have more job options and can even create jobs for others. On the other hand, the benefits of fancy bridges and buildings are mostly enjoyed by people in big cities. The government should cut waste in the construction of physical infrastructure and invest more in education, research, and culture instead.	677
4	Efforts determine success	Equal opportunity: Differences in privilege lead to unequal opportunity	Indonesia claims to support all its citizens, but why are the lives of the rich and the poor so vastly different? From birth, children from poor families face disadvantages. Their mothers often don't get enough healthcare and good food during pregnancy. As they grow up, their families can't afford to send them to good schools, and they often drop out. When they become adults, they struggle to find good jobs, keeping them and their families in poverty. On the other hand, children from wealthy families have a good start from the very beginning. Their mothers are healthy, they live in clean and safe areas, and they can go to school all the way through graduation. As adults, they can get good jobs, which helps them continue to improve their financial situation. The government needs to improve health, education, and social services for everyone so children from both affluent and disadvantaged families have an equal chance to lead successful lives.	680

Treatment	Dominant narrative	New narrative	Complete text of the stimulus (translated)	Sample size
Control		Placebo stimulus	Indonesia, a country comprising approximately 17,000 islands, is the world's largest archipelago. These islands stretch about 5,000 kilometers across the equator. Indonesia is renowned for its diverse array of cultures, languages, and religions. There are over 300 different ethnic groups and 700 regional languages spoken throughout the country. Some popular tourist destinations include Yogyakarta, which boasts ancient temples like Borobudur and Prambanan, and Bali, an island renowned for its stunning scenery that attracts artists and writers alike. Indonesia's natural landscape features beaches with white, gold, silver, pink, and black sand. It also has many volcanoes, including the famous Mount Tambora. In fact, Indonesia has more volcanoes than any other country in the world. One of the most well-known is Krakatau, located between Sumatra and Java. When it erupted in 1883, its ash traveled around the world three times, and it was one of the deadliest eruptions ever, killing over 36,000 people. In the 7th century AD, the Sriwijaya kingdom in Sumatra was a powerful trading center and a center of Buddhist spread. Later, in the 13th century, the Majapahit kingdom in East Java became extremely powerful, controlling most of the islands and leaving behind a rich cultural legacy. Indonesia is also renowned for its traditional handicrafts, performing arts, and diverse cuisine, all of which showcase its rich and varied cultural heritage.	898
Total				3,638

The "No victimization" narrative directly questions the underlying assumption behind the dominant narrative that sacrifice is necessary for growth and development. [Bahana and C4C \(2025\)](#) found that this is a recurring and prominent theme in the development narrative propagated by Indonesian policymakers.

The "Balanced development" narrative seeks to counter the prevailing notion that development is synonymous with infrastructure construction. This common conceptualization among the public has been exploited by Indonesia's past administration as a justification for promoting numerous grand development projects ([Warburton, 2016](#)), despite many questioning their impact on welfare ([Basri, 2017](#); [Tasyarani, 2025](#)). The consequence of this conceptualization is that less physically tangible projects, or even projects that limits physical development, like environmental preservation to ensure certain areas can maintain their ecological services, are less likely to be viewed as development, despite their undeniably important contributions to human welfare ([Bahana and C4C, 2025](#)).

The final "Equal opportunity" narrative aims to mitigate public stigma and enhance public perception of social assistance programs. Social assistance programs are crucial for a just green transition, as workers in environmentally destructive sectors will inevitably be displaced ([UNDP, 2021](#)). One of the most effective tools to help those who lose out in an economic transition to stay afloat is social assistance.

However, this will be difficult to achieve unless the dominant narrative around social assistance changes. [Bahana and C4C \(2025\)](#) found that the public holds a firm belief that people should receive government assistance only if the cause of hardship is external factors and only for a short period. Once the hardship ends, assistance must cease. This opinion aligns with the prevailing narrative on social assistance in Indonesia ([Bimardhika, 2024](#)). Despite robust empirical evidence demonstrating its humongous benefits and its limited distortionary effects, social assistance continues to carry a persistent negative stigma ([Bimardhika, 2022](#)).

Finally, to ensure that the control group has a comparable cognitive load to the treatment group when completing the survey, we assign them to read a placebo stimulus containing general information about Indonesia. We make the placebo stimulus slightly longer than the treatment stimulus because we ask fewer survey questions of the control respondents.¹⁸

3.2.2 Conjoint experiment

In the survey, we also conduct a conjoint experiment that is independent of the main experiment. We are interested in what aspects of development the public values the most. Accordingly, we adapt the conjoint design used by [Marshall et al. \(2023\)](#).

¹⁸For example, we do not ask the control group which paragraph from the stimulus resonates most with them, or whether they agree with the stimulus.

We require participants to choose their preferred development profile from a randomized pair ten times. Each profile has four attributes that characterize the development. The first concerns the distribution of income increases. The second concerns the rights of people displaced by development. The third concerns striking a balance between environmental preservation and development. The final one is about the degree of public participation in development. We present the complete list of attributes and levels in Table C.2.¹⁹

3.2.3 Outcomes

In total, we have seven families of outcomes. While our narratives convey varying messages, several common threads bind them. Our narratives invite readers to rethink how the economy and development should function. As such, we ask our sample what they think should be the nation's development priority and their opinions on a battery of items about the economy. Since the "Fix the distribution" and "Equal opportunity narratives directly address redistribution and economic privilege, we also inquire about whom the government should prioritize in the economy and what determines economic success.

As the final paragraph of each narrative calls for government action, we investigate whether this impacts policy support across a range of items related to social equality and environmental policy ideas. While our narratives do not directly call for citizen action, they might ignite a sense of self-relevance regarding these issues (Sinclair et al., 2025); hence, we investigate the impact on personal efficacy, adapting from the module used by NEON et al. (2018). Finally, we are interested in how narratives can alter the relative importance of various development aspects in a conjoint experiment. We present the complete variable construction in Table B.1 and B.2. Our full questionnaire is also available in our AEA Trial Registry.

4 Empirical approach

4.1 Balance test

We find that the treatment and control groups are similar in characteristics. Following our pre-analysis plan (PAP), we use the omnibus F-test of joint orthogonality method proposed by Kerwin et al. (2024) as our primary method to assess the overall balance between treatment

¹⁹With four attributes, where the first two have three levels, and the other two have two levels, a full factorial design would have 36 profiles. We implement a fractional factorial design using only 15 carefully selected profiles, chosen to maximize profile diversity and balance between attribute levels. We then show each respondent 10 randomly selected profile pairs from a universe of 105 unique possible pairings based on 15 selected profiles.

and control groups across participant characteristics.²⁰

Our test yields an F-statistic of 1.35.²¹ All 9,979 successful simulations out of 10,000 randomized treatment reassignment simulations yield a simulated F-statistic greater than 1.35, indicating a very high p-value. As such, we cannot reject the null hypothesis that there are no overall statistical differences in participant characteristics between treatment and control groups.²²

4.2 Estimation framework

4.2.1 Main experiment

Following our PAP, we estimate several regression models to obtain treatment effects for each outcome in the main experiment. Our basic estimation model is Equation 1, which simply compares each treatment arm to the control group in a regression framework.²³

$$Y_i = \beta_i + \sum_{j=1}^4 \tau_{ij} T_{ij} + \varepsilon_i \quad (1)$$

Y_i is the outcome variable of individual i . $\sum_{j=1}^4 T_{ij}$ is a vector of treatment dummies for individual i . We include four dummy variables corresponding to the four treatment arms to estimate treatment effects for each arm. Each treatment dummy yields our coefficient of interest, τ_{ij} , which is essentially the mean difference of the outcome between treatment arm j and the control group.²⁵

In principle, since treatment assignment is random, regressing the outcome on the treatment indicator, as in Equation 1, should yield an unbiased estimate of the treatment effect. Nonetheless, we run an extension of Model 1 as a sensitivity analysis to ensure the robustness of our results. We express this extension in Equation 2.

$$Y_i = \beta_i + \sum_{j=1}^4 \tau_{ij} T_{ij} + \sum_{k=1}^K \theta_{ik} X_{ik} + \varepsilon_i \quad (2)$$

.

²⁰Table B.1 provides the complete list of participant characteristics used for balance testing, which is also consistent with our PAP. We use MANOVA rather than OLS to account for the multiple-treatment-arm design, as Kerwin et al. (2024) also recommended.

²¹The estimate is the same across four common MANOVA estimands (Wilks' Lambda, Pillai's trace, Lawley-Hotelling trace, and Roy's largest root)

²²As a robustness check, we also provide the conventional balance testing results in Online Appendix OA.2. We find that none of the covariates is imbalanced.

²³²⁴

²⁵ β_i and ε_i are the regression intercept and idiosyncratic errors, respectively.

In the extended model, we estimate the treatment effect while controlling for several covariates to improve precision. $\sum_{k=1}^K X_{ik}$ is a vector of covariates of K numbers.²⁶

Across all models²⁷, we follow the common approach in other closely related studies that transform Likert outcome variables into binary indicators²⁸ and employ OLS as the primary estimator to compute the treatment effect.²⁹

4.2.2 Conjoint experiment

We have three interests in conjoint analysis. The first is to study the relative importance of each development attribute, using the common calculation method by [Rahman and Lorica \(1999\)](#).

The second is to examine the Average Marginal Component Effect (AMCE) for each attribute level.³⁰ We express our estimation model in Equation 3.

$$Y_i = \beta_i + \sum_{l=1}^{10} \delta_{il} C_{il} + \varepsilon_i \quad (3)$$

To run this model, we first set the data into long form where each row corresponds to a respondent-profile pair.³¹ In this setup, Y_i is a binary indicator of whether the respondent selected the corresponding profile within a profile pair. $\sum_{j=1}^{10} C_{ij}$ is a vector of attribute level dummies. δ_{ij} is the AMCE.

The third is to investigate how each relevant treatment arm mediated or intensified the AMCE through interaction effects between each attribute level and its respective relevant treatment arms. We extend Equation 3 to Equation 4 to meet this objective.

²⁶Following [Kerwin \(2025\)](#) suggestion, we plug the participant characteristics used for the balance test into the Post Double Selection Lasso (PDS Lasso) procedure developed by [Chernozhukov et al. \(2016\)](#) and let the procedure determine the appropriate control variables for each outcome.

²⁷We use the [White \(1980\)](#) robust sandwich heteroskedasticity consistent estimator to compute the standard errors.

²⁸See [Dechezleprêtre et al. \(2025\)](#); [Sim et al. \(2024\)](#). The full description of the transformations is available in [B.2](#).

²⁹See [Friedman \(2012\)](#); [Bloem \(2019\)](#); [Gibson \(2019\)](#) on why OLS is an appropriate estimator for binary outcome variables. As a robustness check, we also perform ordered logit regressions on the untransformed Likert measure as the dependent variable, and present the results in Online Appendix OA.4. This exercise allows us to observe changes in views along the discrete scale as a result of the treatment. This also helps us assess whether the OLS results are biased by the way we transform Likert-scale outcomes into dummy variables. We find that the ordered logit results are largely consistent with the OLS results.

³⁰AMCE is the relative preference of each attribute level against the base level in the attribute. We indicate which level is the base level for each attribute in [Table C.2](#). AMCE is independent of the main experiment due to randomization.

³¹Each profile is identified by its attribute level dummy columns. Given that each respondent completes 10 paired profile selection task, each respondent has 20 rows in this data format. As is standard in a conjoint experiment involving paired profile selection, we employ OLS as estimator and cluster the standard errors at the respondent ID.

$$Y_i = \beta_i + \sum_{l=1}^{10} \delta_{il} C_{il} + \sum_{j=1}^4 \tau_{ij} T_{ij} + \sum_{m=1}^M \rho_{im} C_{il} T_{ij} + \varepsilon_i \quad (4)$$

The only difference with the basic model is $\sum_{m=1}^M C_{il} T_{ij}$, which is a vector of relevant treatment and attribute level interactions. ρ_{im} tells us whether and how the treatment moderates AMCE.

4.3 Hypothesis testing setup

We ground our hypothesis testing in theory and intuition. As such, while we ask the same questions to all respondents to equalize cognitive burden across study arms, we only estimate the treatment effect on the relevant outcome for each treatment arm. We determine the relevant outcome for each treatment based on the stimulus content. If any aspect of the stimulus may affect the outcome, either positively or negatively, we consider the outcome relevant to the treatment arm.

We present our complete regression plan in Appendix C. In Table C.1, we detail the mapping of each outcome from the main experiment onto its relevant treatment arms, grouped by the family of outcomes we use for multiple hypothesis testing. The final two columns provide information about the estimator we use to run the models. In Table C.2, we detail the relevant interaction effects (treatment-moderating or treatment-enhancing) we want to examine in the conjoint experiment. Each treatment column indicates which treatment arm interacts with each attribute level in the row.

As we promised in our PAP, given the multitude of outcomes and multiple arms design, we adjust every regression for multiple hypothesis testing. To account for the correlation among outcomes in a family in the analysis of the main experiment, we employ the [Westfall and Young \(1993\)](#) p-value adjustment, as implemented in the statistical program developed by [Jones et al. \(2019\)](#). For the conjoint analysis, we apply the p-value adjustment of [Holm \(1979\)](#) because we have only a single outcome variable (whether the profile was chosen) and numerous independent variables.³²

³²Both [Westfall and Young \(1993\)](#) and [Holm \(1979\)](#) procedures control the family-wise error rate (FWER). This is a more stringent approach than the alternative, the false discovery rate (FDR), for adjusting multiple hypothesis testing ([Coppock, 2021](#)). This makes FWER more suitable for a causal analysis that seeks to minimize false positives ([Lindquist and Mejia, 2015](#)).

5 Results

5.1 Main experiment

5.1.1 Treatment compliance

We consider participants to be complying with treatment when they read and accurately interpret the narrative’s key message. The latter criterion is crucial because, in theory, narratives can lead to belief updating when participants understand their intended meaning. To assess this, we ask respondents to identify the narrative’s key message after reading the stimulus.³³

Figure D.1 compares the word count in the original Indonesian text and the percentage of correct interpretation of the narrative’s key message across the four stimuli. The figure indicates comprehension is optimized around 75-100 words. Once the narrative script exceeds 100 words, comprehension drops dramatically. The first three narratives are well understood by respondents, with more than 80% correctly interpreting their key messages. In contrast, fewer than half of the respondents assigned to reading the “Equal opportunity” narrative interpret it correctly. This may be attributable to its substantially higher word count, which can increase cognitive load or dilute the central message.

Considering this, while we primarily examine the treatment’s intent-to-treat (ITT) effect³⁴, we nonetheless estimate the treatment-on-treated (TOT) effect to evaluate whether imperfect compliance distorts our main findings. We relegate the explanation of the TOT estimation framework and its results to Online Appendix OA.5. We find that the TOT results closely mirror the ITT results.

5.1.2 Development priority

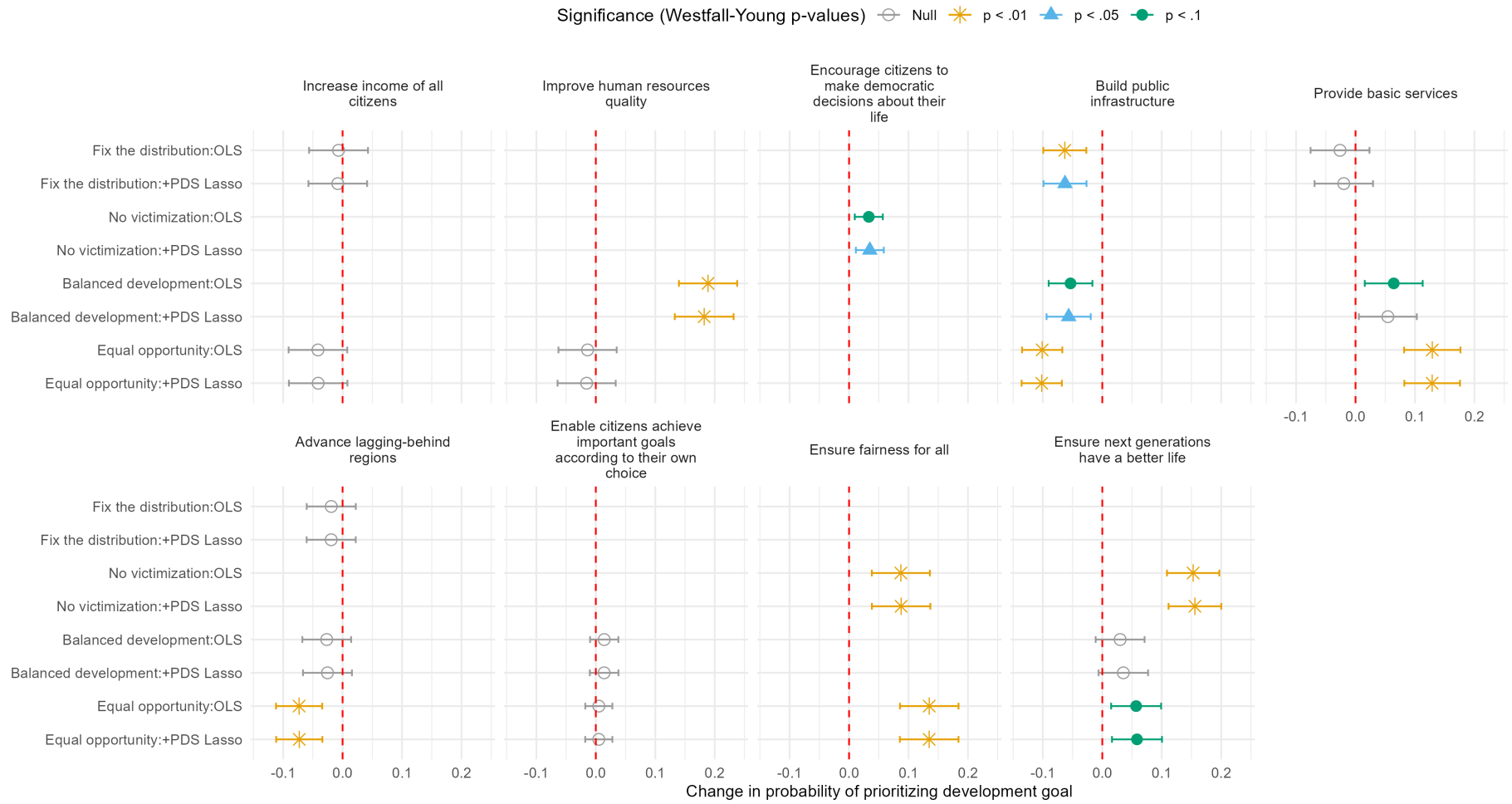
We ask each respondent to choose up to 3 items they believe should be the government’s development goals from a list of 12 possible options.³⁵ The strict multiple-response cap means that respondents have to weigh which goal to include in their top three, essentially making the answer their development priority. Figure 1 shows the regression results for nine stimulus-relevant outcomes, with 95% confidence intervals and Westfall-Young statistical significance.

³³We check for compliance further in Online Appendix OA.6. We find that stimulus reading time tends to correlate positively with the length of the stimulus, and respondents with correct stimulus interpretation tend to read longer than those who don’t (albeit the difference is not statistically significant). Both suggest that respondents indeed read the stimulus.

³⁴It is the common approach in randomized experiments with imperfect compliance (Angrist and Pischke, 2009)

³⁵We provide both stimulus-relevant and stimulus-irrelevant options to avoid restricting responses to only stimulus-relevant ones. Given a long list of answer options, we randomize their order in the survey to prevent systematic bias in responses.

Figure 1: Treatment effects on beliefs about development priority



The treatment effects of "Fix the distribution" narrative, which calls for fixing the distribution of development benefits, are largely moot, if not counterproductive. It does not increase the likelihood of prioritizing income improvement for all citizens, providing equitable and quality basic services (such as education, health, and social assistance), and advancing lagging-behind regions. More puzzlingly, it also reduces the likelihood of prioritizing the construction of public infrastructure and facilities by around 6 percentage points.

All the treatment effects of the "No victimization" narrative, which calls for safeguarding the development process from sacrificing people and the environment, appear as expected. It increases the likelihood of prioritizing democratic decision-making by 3 percentage points, ensuring fairness for all citizens in rights, opportunities, and the law by 8 to 9 percentage points, and ensuring that the next generation has a better life by an impressive 15 percentage points. All three are essential to a sustainable and equitable economy.

The "Balanced development" narrative, which emphasizes the balance between human and physical development, yields mixed results. It increases the likelihood of prioritizing the improvement of human resource quality by an astonishing 18 to 19 percentage points and providing equitable and quality basic services by 5 to 6 percentage points, although the latter is not robust in the second model.

Yet, "Balanced development" also has an unintended consequence. It results in a 5 percentage-point reduction in the probability of prioritizing the construction of public infrastructure and facilities. This might be attributed to the sentence "*The government should reduce waste in building physical facilities to increase the budget for education, research, and culture.*" in the script. However, we had expected to see a null effect rather than a statistically significant negative effect here.³⁶

The "Equal opportunity" narrative, which advocates for equalizing opportunities between the rich and the poor, has both positive and negative results. It increases the likelihood of prioritizing the provision of basic services by 12 to 13 percentage points, ensuring fairness for all in rights, opportunities, and the law by 13 percentage points, and ensuring the next generation has a better life by around 5 to 6 percentage points.

Unfortunately, it does not influence the likelihood of prioritizing other goals, such as increasing the income of all citizens, improving the quality of human resources, and enabling citizens to pursue their goals, all of which are related to equal opportunities. Worse, it reduces

³⁶"Balanced development" also has some null effects. It does not affect the probability of prioritizing the advancement of underdeveloped regions, despite the presence of an explicit statement of "*the benefits of flyovers and magnificent buildings are only enjoyed by people in big cities*" in the stimulus. It also does not affect the probability of prioritizing enabling citizens to achieve the important goals they set for themselves and ensuring the next generation has a better life, even though both are essential to human development. The former might be related to the collectivist culture prevalent in Indonesian society (Hoon, 2004).

the likelihood of prioritizing the construction of public infrastructure and facilities by 10 percentage points and of advancing underdeveloped regions by 7 percentage points.

5.1.3 Opinions about the economy

We ask the respondents to indicate their level of agreement on six statements related to a vision of an equitable and sustainable economy. Figure 2 visualizes the regression results with 95% confidence intervals, and Westfall-Young corrected statistical significance.

The figure shows that none of the narratives has a robust impact on the first two statements: 1) “If people’s economic conditions are good, we can reduce dependence on economic activities that damage or pollute the environment” and 2) “The government may relocate people in order to build public facilities for the public interest, provided that the government provides replacement housing”. Meanwhile, the “No victimization” narrative increases the probability of agreeing with “Citizens should not just accept the government decisions without question” by 5 to 6 percentage points.

The “Fix the distribution” and “Equal opportunity” narratives reduce the likelihood of agreeing that wealth inequality is normal by 8 and 9 percentage points, respectively. Both narratives also increase the likelihood of agreeing that the government’s decisions determine who rules the economy by 10 and 7 percentage points, respectively. Neither narrative increases the likelihood of agreeing that the government’s laws and rules determine the chances of people succeeding.

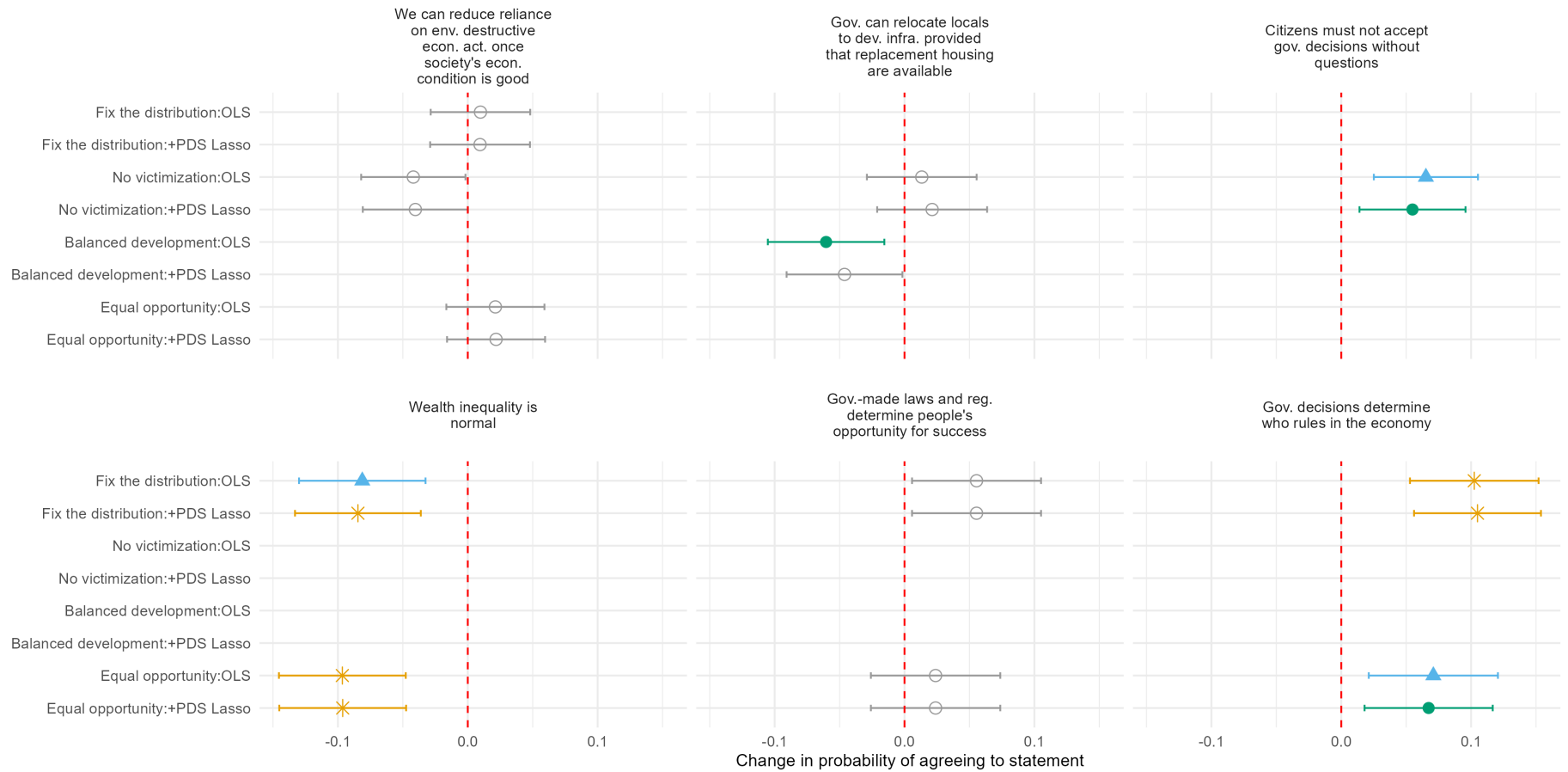
To further explore their beliefs about the economy, we also collect two additional outcomes. First, we ask respondents to rank eight types of people in terms of whom the government should prioritize during rapid economic growth.³⁷ Second, we ask respondents to rate four key factors based on how much they play a role in the economy.³⁸ We present the regression results with 95% confidence intervals and Westfall-Young corrected statistical significance in Figures D.2 and D.3, respectively. Neither “Fix the distribution” nor “Equal opportunity” narrative has any impact on these beliefs.

³⁷To prevent systematic bias, we randomize the order of answer options. We hope that “Fix the distribution” and “Equal opportunity” might persuade the public that poor individuals and those in lagging regions need government support most during periods of rapid economic growth, enabling them to share in economic benefits like everyone else.

³⁸We suspect that “Fix the distribution” and “Equal opportunity” narratives may convince people that laws and regulations play a sizeable role in determining the state of the economy, while fate or human nature less so. This belief serves as a crucial foundation for an equitable economy.

Figure 2: Treatment effects on opinions about the economy

Significance (Westfall-Young p-values) ○ Null * $p < .01$ ▲ $p < .05$ ● $p < .1$



5.1.4 Policy support

We ask respondents to indicate their level of support for 13 policy ideas that are relevant to building a sustainable and just economy.³⁹ Figures 3 and 4 visualize the regression results with 95% confidence intervals and Westfall-Young corrected statistical significance.⁴⁰

As in the effect on development priority, the effects of the “Fix the distribution” narrative are again mostly null, if not negative. Some of the null effects might be good news. It basically means that the narrative does not lead to unintended effects on policy ideas crucial to the energy transition agenda, yet might attract opposition, such as: 1) Fuel and electricity price hikes accompanied by cash transfers to all, not just the poor, and 2) Higher motor vehicle taxes.

However, null effects also mean the narrative is unable to engender progress on essential social equality policy ideas, such as: 1) The government providing financial assistance to people who have recently lost their jobs (unemployment insurance), or 2) Broader coverage of social assistance recipients, including the middle class. It has no impact on supporting a larger amount of social assistance, which is the most basic form of redistribution.⁴¹ Worse, the narrative unfortunately reduces the likelihood of supporting “Subsidies for energy-efficient homes and buildings” (Figure 3) by 7 percentage points.

While we have shown it to be promising in driving key development agendas, “No victimization” has virtually no robust impact on any policy idea. We expect the narrative to convince people to endorse environmentally friendly policies, given its explicit emphasis on preventing development from sacrificing the environment. The unadjusted regressions suggest that the narrative reduces support for forest clearance for agricultural and infrastructure purposes, but these effects are not sustained after adjusting for multiple hypotheses.

“Balanced development” has some mixed impacts. First, the narrative has no impact on the following policy ideas: 1) A larger government budget for the development of green technology and renewable energy, 2) Residents who are displaced due to infrastructure development projects benefit from the utilization of the infrastructure, and 3) Subsidize industries to switch to green technology and energy.

Most disappointingly, “Balanced development” reduces support towards: 1) A larger government budget for the development of a mass, integrated, and environmentally friendly public transportation system, and 2) Subsidies for energy-efficient homes and buildings by 3 to 5 and 6 percentage points, respectively.

³⁹We randomize the order of item appearance in the survey to prevent systematic bias from item ordering.

⁴⁰As the column headers in the figures only display the shortened version of the policy statement, please consult Table B.2 or C.1 for the full policy statement.

⁴¹This suggests that the public may not perceive social protection as redistribution but rather as temporary assistance (Bimardhika, 2024), consistent with our earlier study (Bahana and C4C, 2025).

Figure 3: Treatment effects on policy support

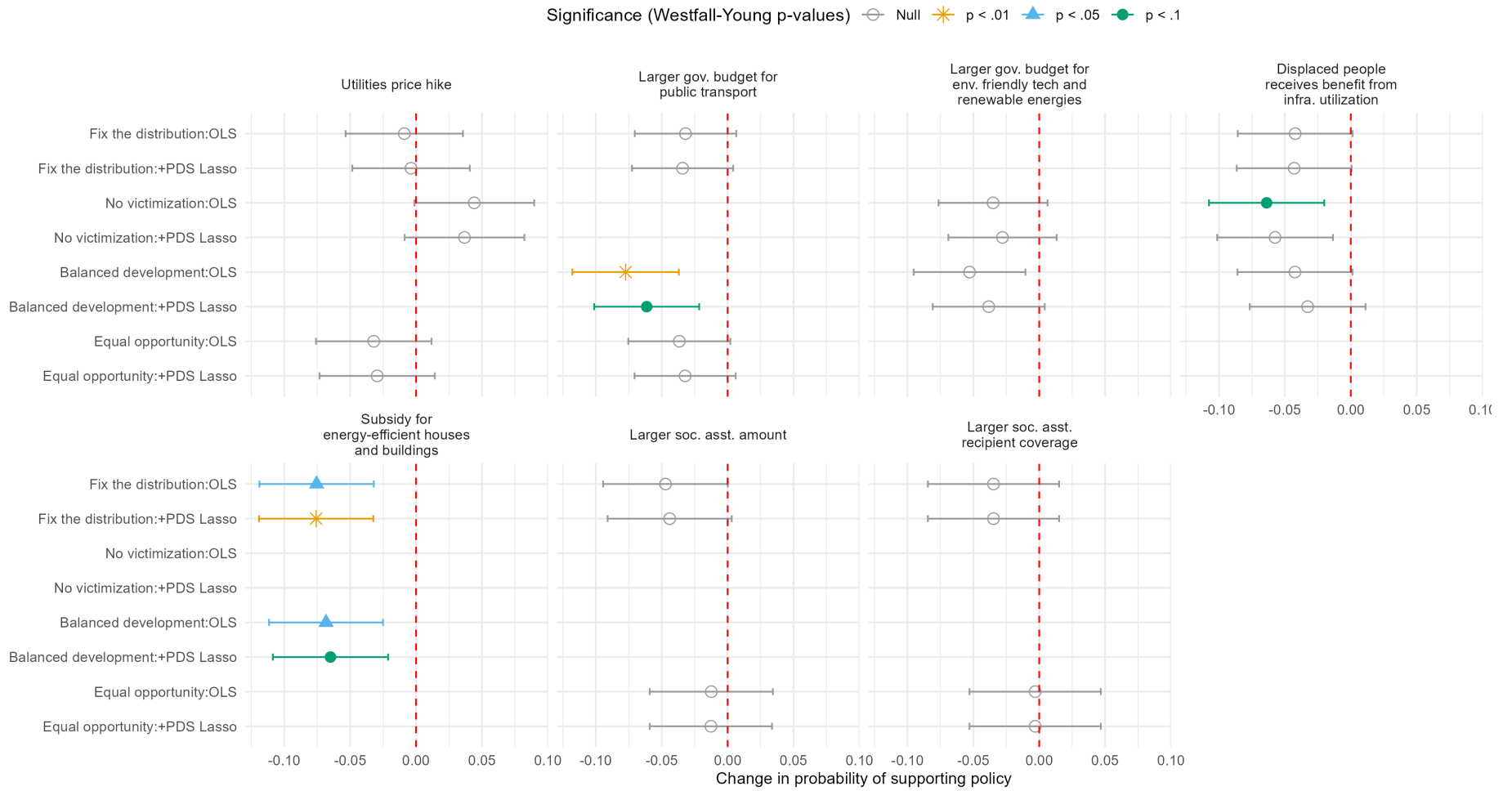
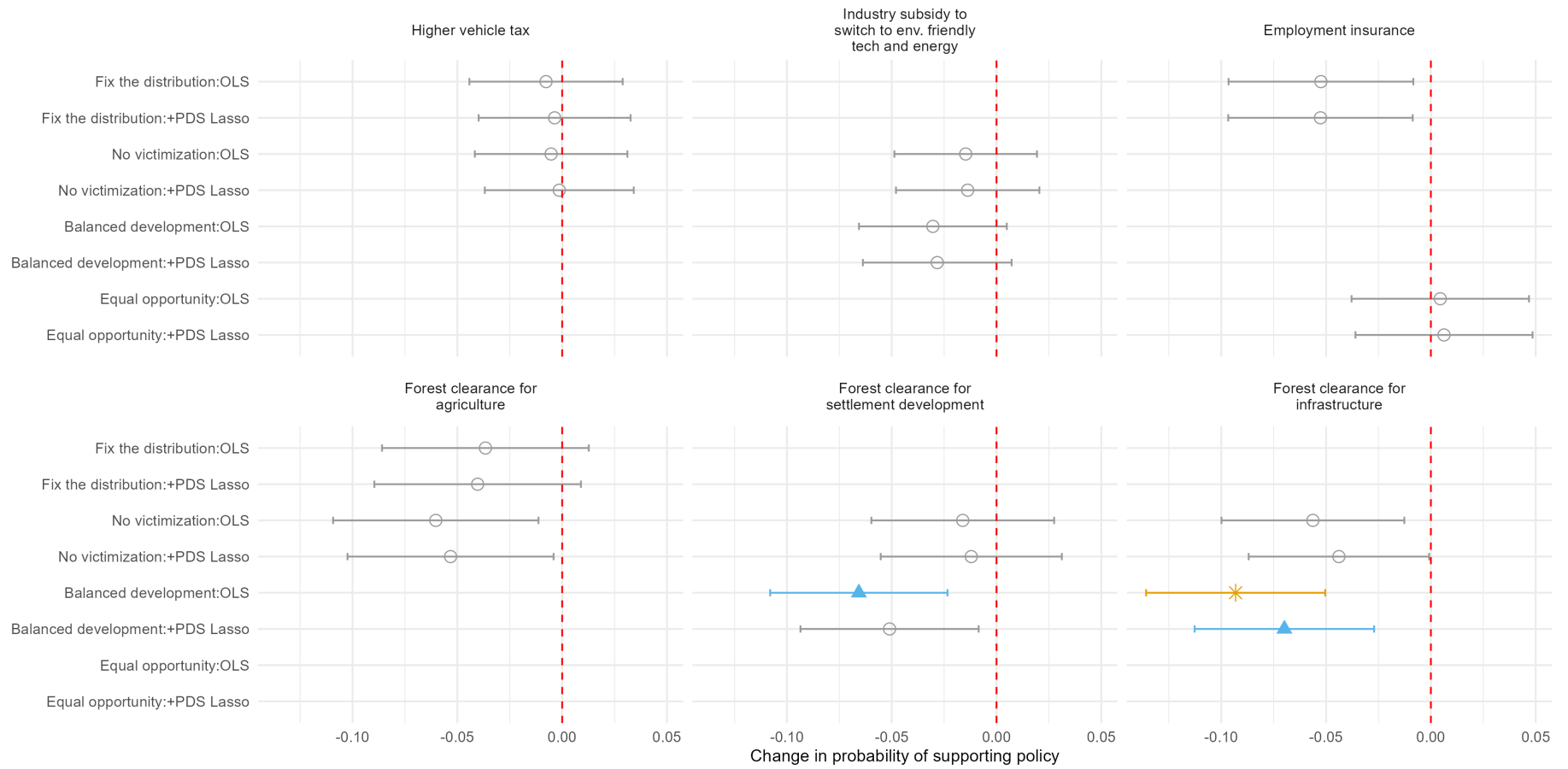


Figure 4: Treatment effects on policy support

Significance (Westfall-Young p-values) ○ Null ✱ $p < .01$ ▲ $p < .05$



Nevertheless, the good news is that "Balanced development" reduces support for cutting down forests for settlement and infrastructure development by 5 to 6 percentage points and 6 to 9 percentage points, respectively, although the former is not robust across models.

The "Equal opportunity" closely mirrors the results of "Fix the distribution". Since they share a similar theme, the outcome set is also similar. This narrative is largely ineffective in stimulating changes in policy support.

5.1.5 Personal efficacy

The main experiment's final outcome is personal efficacy. We ask respondents five questions about their beliefs regarding personal and collective responsibility for creating an equitable and inclusive economy.⁴² Figure D.4 summarizes the regression results with 95% confidence intervals and Westfall-Young corrected statistical significance. We find that neither the "Fix the distribution" nor the "Equal opportunity" narratives are effective at provoking personal and collective efficacy.

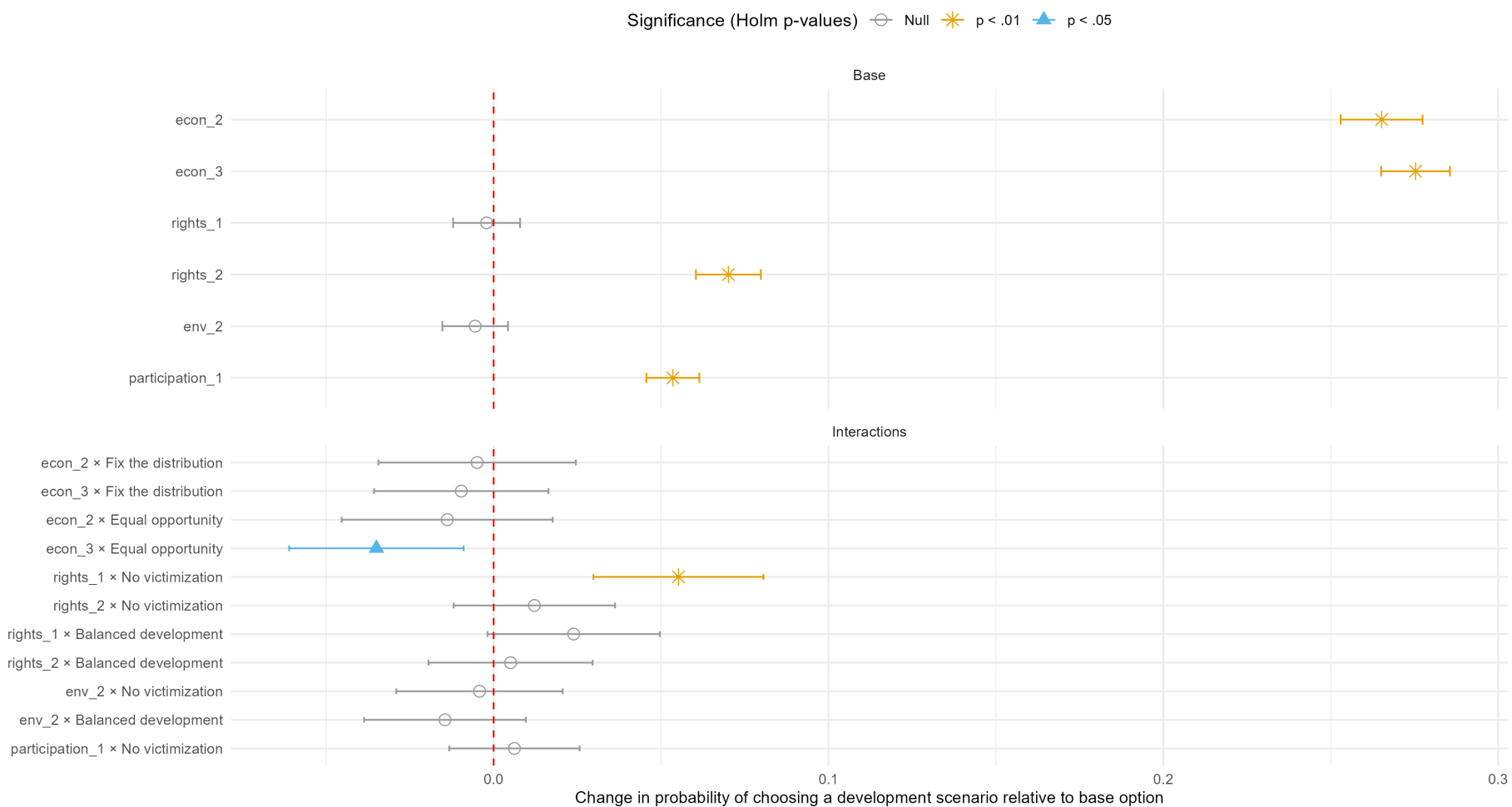
The results for personal and collective efficacy appear to align with the emotions evoked by the stimulus and with the literature on the role of emotions in driving social change. Online Appendix OA.1.3 shows that sadness is the most dominant emotion across stimuli. Although the evidence for the magnitude of impact from each type of emotion is still mixed (Çakmak et al., 2024), studies have found that sadness is less potent in driving social change compared to hope (Greenaway et al., 2016), while anger is noted as a key emotion for mobilizing activism (Ransan-Cooper et al., 2018).

5.2 Conjoint experiment

Figure D.5 illustrates the distribution of decision weights when respondents select their preferred development profile. The distribution of income increase remains the most important factor in people's preferences regarding the development profile. Meanwhile, respondents appear indifferent to environmental protection concerns. Over two-thirds of respondents' consideration centers on the specific feature of the improvement of the economic condition (ECON) attribute. The remaining consideration weight is distributed across the other three attributes, with environmental protection (ENV) accounting for less than 2 percent of the total. This echoes the findings of our earlier study that environmental protection is barely part of the Indonesian public's conceptualization of what makes a successful development. (Bahana and C4C, 2025).

⁴²We randomize the order of the questions to prevent systematic bias due to ordering.

Figure 5: Conjoint experiment results



We now inspect the AMCE of each attribute level. Figure 5 visualizes the regression results with 95% confidence intervals and Holm corrected statistical significance. We split the results into two panels. The "Base" panel shows the AMCE from Equation 3. The "Interactions" panel shows the narrative treatment effect on the preference over attribute levels from Equation 4.

We find that ECON3 has the largest positive AMCE. ECON2 follows closely. When either feature is present in the development profile, it increases the likelihood of profile selection by more than 26 percentage points compared to ECON1. This means people dislike a development in which income increases are concentrated among the rich (ECON1). Yet, the slightly higher AMCE for ECON3 (modest income increase for everyone) over ECON2 (higher income increase for poor and near-poor and lower for middle- and upper-income groups) is also interesting, suggesting that the Indonesian public does not have a strong preference for growth that benefits everyone equitably, and is amenable to one that benefits everyone equally.

There is a notable gap between ECON2 and RIGHTS2, the next highest positive AMCE. RIGHTS2 increases the likelihood of profile selection by seven percentage points compared to RIGHTS3. This suggests that people cannot simply accept displacement for development, even when compensation is involved. People prefer that citizens also benefit financially from development. Nonetheless, the AMCE is nowhere close to the ECON's. Meanwhile, RIGHTS1 does not affect the probability of profile selection.

Respondents are completely indifferent to the development's environmental attribute, as ENV2 is not statistically significant. Finally, PARTICIPATION1 increases the likelihood of profile selection by 5 percentage points. This suggests that people still desire citizen participation in the development process. This is consistent with their preference on RIGHTS2.

There are two statistically significant estimates in the interaction models. First, the "Equal opportunity" narrative reduces preference for ECON3 by 3 percentage points. This means people exposed to this narrative are less likely to select a development profile in which the income gain feature is equal across income groups, albeit modest. However, this does not mean they now prefer ECON2, where the income gain is higher for the lower-income group and lower for the middle- and higher-income groups, as the treatment effect at this attribute level is not significant.

The "No victimization" narrative increases the probability of selecting a development profile with a RIGHTS1 feature (No one should be displaced, even if they receive compensation) by 5 percentage points. This is understandable given the narrative's explicit focus on not sacrificing the people for development.

6 Discussions and implications

We learn that even a well-intentioned narrative can lead to unintended consequences. Let us now try to better understand the driver behind these puzzling results.⁴³ We suspect that this backlash might be correlated with the most resonating paragraph from each stimulus. To investigate this, we examine the relationship between the most resonating paragraph and the outcome that reveals the backlash.⁴⁴

6.1 Fix the distribution

The "Fix the distribution" narrative, which calls for fixing the distribution of development spoils, has notable unintended effects. It reduces prioritization of public infrastructure and facilities, and it undermines support for the policy idea of subsidizing energy-efficient homes and buildings.

Figure D.6 indicate that the likelihood of supporting subsidies for energy-efficient homes and buildings tends to fall when the respondents choose the paragraph "*All this time, the benefits of development have not reached all citizens. Those who enjoy it are the rich and powerful. Those at the bottom are still not prosperous.*" as the most resonating. This seems to imply that people associate energy-efficient homes and buildings (which count as infrastructure) with an extravagance that only the rich can afford.

6.2 No victimization

There is no backlash for the "No victimization" narrative. However, the narrative is unable to drive any support for progressive policy ideas. We posit that the vague call to action (CTA) in the script's final paragraph might have undermined the narrative's effectiveness. While the CTAs of the other narratives are quite specific about what the government should do (redistribute better, cut waste, expand coverage), the CTA of this narrative is more focused on asking the government to adhere to a certain development principle and is unclear about the specific action. We speculate that making the CTA of this narrative more specific and concrete might yield impactful policy support.

⁴³To study the respondents' immediate reaction to the stimulus further, refer to Online Appendix OA.1, which provides an in-depth inspection of three immediate reactions to the stimulus: which paragraph respondents find most resonant, to what extent they agree with the paragraph, and the emotional reactions they experience after reading the stimulus.

⁴⁴Note that this interpretation is merely correlational and not causal. The selection of the paragraph as the most resonating is not random.

6.3 Balanced development

As in "Fix the distribution", "Balanced development" narrative, which advocates a balanced focus on human and infrastructure development, also lowers prioritization of building public infrastructure and facilities, and reduces support for key policy ideas essential for energy transition: 1) A larger government budget for the development of a mass, integrated, and environmentally friendly public transportation system, and 2) Subsidies for energy-efficient homes and buildings.

In Figure D.7, we find that the likelihood of supporting these policies falls when the respondent chooses the paragraph "*The government should reduce waste in building physical facilities to increase the budget for education, research, and culture.*" as the most resonating, especially among the compliers (those who accurately interpreted the message). Associating "building physical facilities" with "waste" in the script might give the impression that environmentally friendly public transport systems and energy-efficient buildings are wasteful or non-essential.

The lavish and extravagant impression of energy transition projects might be exaggerated by the limited public literacy of renewable energy and climate change⁴⁵, and the populist narrative that there is a tradeoff between environmental action and immediate economic fulfillment (C4C, 2021, 2022).

As in "No victimization", revising the CTA of this narrative might help ameliorate the problem. One might reframe the CTA as infrastructure development efficiency or more effective infrastructure spending, rather than associating infrastructure spending with wastefulness.

6.4 Equal opportunity

The "Equal opportunity" narrative yields some perplexing results. In terms of development goals, it increases prioritization for ensuring fairness for all, which aligns with the increased emphasis on providing equitable basic services. However, it is inconsistent with reduced prioritization on advancing lagging-behind regions.

We can better understand this result by linking it to paragraph resonance (Figure OA.4 in the Online Appendix) and agreement (Figure OA.8 in the Online Appendix). The paragraphs about the privilege gap (2nd and 3rd paragraphs of the narrative script) are the least selected as the most resonant. Yet, it draws the most disagreement. In addition, the selection of the most resonant paragraph in this experiment arm is heavily concentrated in the final para-

⁴⁵Our earlier study (C4C, 2021) found that only 23% of urban youth had heard the term "renewable energy" before being interviewed. Additionally, only 50% believes that Indonesia's oil and coal reserves would eventually deplete.

graph, which mandates that the government provide basic services to equalize opportunities.

It seems that people prefer to redirect the discourse on inequality of opportunity away from stark differences in privilege toward what the government can do to help everyone equally. As such, prioritizing certain groups, such as those living in disadvantaged regions, over others may not be favorable. This is consistent with the conjoint analysis, which finds that the Indonesian public appears to believe that everyone is entitled to benefit equally (albeit unequitably), regardless of their privileges or lack thereof.

In Figure D.8, we also show that complying participants in this treatment are less likely to prioritize advancing underdeveloped regions when they choose the paragraph highlighting the privilege of rich children as the most resonant. As prioritizing the development of underdeveloped regions is a form of redistribution, respondents' disagreement with the privileges of rich children might indicate they do not see redistribution as necessary. As such, exposing people to their privileges appears to be an ineffective, if not counterproductive, approach to achieving social justice goals.

Several other studies strengthen our inference. [Hoy et al. \(2024\)](#) found that informing Indonesians of their true relative position in the socioeconomic distribution reduces support for redistribution among richer respondents. This insight is especially relevant to support our interpretation, given that our sample is more urban and educated than the representative national population. The same study also found that Indonesians prefer job creation over redistribution as a means to combat inequality.

Our insights also corroborate those of [Koo et al. \(2023\)](#), who found that wealthy individuals who grew up poor tend to be less sensitive to the difficulties faced by the poor and less supportive of redistribution, as they are less likely to believe that social mobility is challenging. This insight also sheds some light on the null effects of the narratives on social protection policies.

Critics of our interpretation might argue that the failure of the stark privilege framing might simply be driven by our more urban and educated sample. However, another study has provided evidence that such framing may not work for those who are less well-off as well. [Hoy and Mager \(2021\)](#) demonstrated that individuals who are informed that they are relatively poorer than they initially thought are less likely to be concerned about inequality and are not more supportive of redistribution, as people tend to benchmark their level of welfare to the people around them.⁴⁶

We suspect that this aversion to discussing privilege might stem from the “just world” be-

⁴⁶In line with this, [Arsendy \(2020\)](#) highlighted that Indonesians tend to live in socially segregated clusters, with people within these clusters being predominantly homogeneous. As such, they mostly interact with people of the same social class.

lief ([Lerner and Montada, 1998](#)). This cognitive bias leads people to believe the world is fundamentally fair and that people get what they work for. As empirical support, [Almås et al. \(2025\)](#) found that Indonesians are more likely to believe that merit plays a larger role than luck in shaping inequality. The same study also documented that the dominant type of fairness in Indonesia is “libertarian,” in which people believe it is fair for workers to receive their original earnings and thus do not redistribute at all, regardless of the source of inequality (luck or productivity).

Furthermore, [Trager and Atari \(2025\)](#) found that societies with a strong respect for proportionality, which shares a similar worldview to “just world” belief ([Atari et al., 2023](#)), such as Indonesia, which we verified in a previous study ([LabNarasi, 2026](#)), are more likely to tolerate wealth accumulation. This conceptualization of fairness has significant implications for promoting tax and social protection reforms.

However, the effects the “Equal opportunity” narrative has on opinions about the economy seem to contest this proposition. The stimulus makes the participants less likely to agree that wealth inequality is a normal phenomenon. While bewildering, another study actually also found that Indonesian’s belief in meritocracy tends to weaken after receiving information about the state of high income inequality and low social mobility in the country ([Mijis and Hoy, 2022](#)).

To explain this paradox, we surmise that it might be easier for those in privileged socioeconomic positions to accept that external factors influence individuals’ socioeconomic achievement than to acknowledge some degree of undeservedness in their success.⁴⁷ [Arsendy \(2021\)](#) has noted how people eventually resort to mentioning luck as a justification for their success, rather than acknowledging their privileges.

Ultimately, the key message about equal opportunity remains appealing; however, we need a more effective framing when discussing social justice with the Indonesian public, one that takes into account how the Indonesian society conceptualizes “fairness”.

6.5 Implications for messaging

Our key takeaway is that caution and care are advised when designing narratives, as they might lead to unintended consequences. Among the four narratives, we deem the “No victimization” and “Balanced development” narratives to be the most favorable, provided we make some alterations to the narrative script. We recommend utilizing these two narratives in campaign efforts to attract public support for an equitable and sustainable economy.

⁴⁷Most Indonesians are Muslims, and 98% of Indonesians believe that religion plays a vital part in their life ([Tamir et al., 2020](#)). One of the most popular aphorisms in Islam is “Man proposes, God disposes”, and we speculate that accepting this undeservingness may run contrary to the saying and trigger discomfort.

Furthermore, the contrast between the impact of the “Balanced development” narrative on forest and land-use policies and on energy transition policies confirms that information and narratives are complementary. Deforestation problem is more salient among Indonesians (Leiserowitz et al., 2023). Thus, Indonesians are more able to imagine and connect it to what the narrative preaches. This suggests that while Indonesians may understand the danger of deforestation, they may not grasp the urgency of the energy transition. We posit that this is why we observe statistically significant effects of reduced support on destructive land-use policies (forest clearance), but no meaningful impact, or even adverse effects, on energy transition policies.

While information provision alone may not be sufficient to shift attitudes and beliefs, as the public may lack the necessary lens to interpret it, information deficiency may also hinder the effectiveness of narratives or create framing that can be counterproductive. This means that basic information interventions aimed at improving public literacy of renewable energy, the energy transition, and the climate crisis are still necessary, alongside narrative-based campaigns. Therefore, rather than discarding this narrative, we recommend being more careful and cautious about wordings and information vacuum among the public when using this kind of messaging.

7 Conclusion

This study provides one of the first randomized assessments of broad economic development narratives in a middle-income, rapidly growing economy. By experimentally exposing over 3,600 Indonesian adults to four alternative narratives, (1) equitable distribution of development benefits, (2) safeguarding people and the environment, (3) balancing human and physical development, and (4) equalizing opportunities, we test how narratives can shift public perceptions of development priorities, economic beliefs, and policy support. This study provides new evidence on the kinds of narratives that civil society organizations (CSOs) and think tanks can use in their campaign and policy advocacy work. While many think tanks and CSOs have expended considerable effort to shift the economic narrative in Indonesia, we are still a long way from achieving the vision of an equitable and sustainable economy. Past studies and attempts have shown that information provision alone may not be sufficient to shift attitudes and beliefs. They need to be framed in the direction that reformers desire. Narratives help form that frame in people’s minds.

For policy advocates and CSOs, these results underscore the importance of narrative design: messages that emphasize concrete protections for people and the environment are both effective and safe, while those dwelling on inequality of opportunity or distributional gaps

must be crafted with caution to avoid unintended resistance. Complementing narrative framing with accessible information and repeated exposure is likely necessary to sustain attitude change and build momentum for reform.

Our study has limitations. The online, quota-based sample overrepresents urban, slightly more educated Indonesians, and the short-term measurement cannot capture enduring shifts in beliefs or behavior. Future research should test refined narratives among more diverse populations, explore more sophisticated mediums to deliver the narratives, and examine the cumulative effects of repeated storytelling. By advancing the experimental study of development narratives, this work lays the groundwork for more strategic communication campaigns that align Indonesia's growth story with social justice and environmental sustainability.

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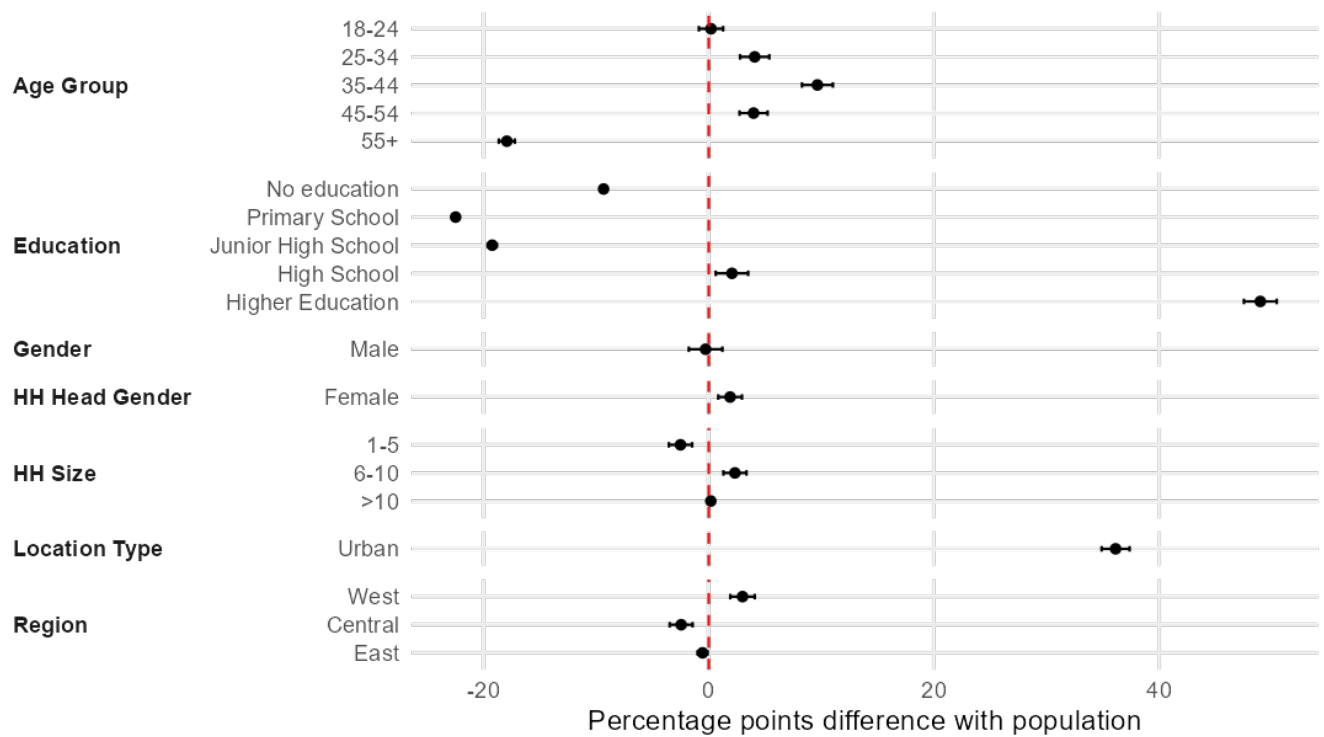
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Appendices

A Appendix A: Sample characteristics

Figure A.1: Statistical difference of key characteristics between the sample and the population



Notes: population data is measured using Susenas (March 2023)

B Appendix B: Variable construction

Table B.1: Basic characteristics and treatment indicators construction

Variable	Description
Basic characteristics	
Region1	An indicator variable equal to 1 if the unit lives in the Western Indonesia Time zone
Region2	An indicator variable equal to 1 if the unit lives in the Central Indonesia Time zone
Region3	An indicator variable equal to 1 if the unit lives in the Eastern Indonesia Time zone
Urban	An indicator variable equal to 1 if the unit lives in an urban area
Male	An indicator variable equal to 1 if the unit is male
Age	Unit's age (in years, discrete, untransformed from raw data)
Unmarried	An indicator variable equal to 1 if the unit is unmarried
Edu1	An indicator variable equal to 1 if the unit's educational attainment is less than the primary level
Edu2	An indicator variable equal to 1 if the unit's educational attainment is at the primary level
Edu3	An indicator variable equal to 1 if the unit's educational attainment is junior secondary level
Edu4	An indicator variable equal to 1 if the unit's educational attainment is senior secondary level
Edu5	An indicator variable equal to 1 if the unit's educational attainment is tertiary level
HH head female	An indicator variable equal to 1 if the unit's household head is female
No social assistance	An indicator variable equal to 1 if the unit's household never receives social assistance
HH size	The total number of individuals living in the unit's household
Social desirability index	Please see Reynolds (1982) for detailed guidance
Treatment indicators	
Treat1	An indicator variable equal to 1 if the unit was randomly assigned to the treatment arm "Fix the distribution"
Treat2	An indicator variable equal to 1 if the unit was randomly assigned to the treatment arm "No victimization"
Treat3	An indicator variable equal to 1 if the unit was randomly assigned to the treatment arm "Balanced development"
Treat4	An indicator variable equal to 1 if the unit was randomly assigned to the treatment arm "Equal opportunity"

Table B.2: Outcome variables construction

Development priority

For each item, an indicator variable equals 1 if the respondent thinks that the item ought to be the government's development goals.

1. Increase the income or economic capability of all citizens
2. Advancing the quality of human resources
3. Encouraging people to make decisions about their lives democratically
4. Erecting infrastructure and facilities for public use
5. Provide equitable and quality basic services (education, health, social assistance)
6. Advancing underdeveloped regions
7. Enable Indonesians to achieve the important things of their choice
8. Ensure fairness for all citizens in rights, opportunities, and the law
9. Ensuring the next generation has a better life

Opinions about the economy

For each statement, an indicator variable equals 1 if the respondent choose "agree" or "strongly agree" to the statement.

1. If people's economic conditions are good, we can reduce dependence on economic activities that damage or pollute the environment.
2. The government may relocate people in order to build public facilities for the public interest (e.g., roads, reservoirs, power plants), provided that the government provides replacement housing.
3. We, as citizens, should not just accept the government's decision.
4. It is natural that there are differences in wealth within society.
5. Laws and rules made by the government determine the chances of people succeeding.
6. The choices the government makes determine who is in power in our economy.

Who should the government prioritize?

For each type of people, an indicator variable equals 1 if the respondent ranks that type among the top three most deserving of government support during rapid economic growth.

1. People with capital
2. People in power
3. People with extensive connections
4. People who work diligently
5. People who are good at seizing opportunities
6. People who don't give in to circumstances
7. People who are poor or underprivileged
8. People living in underdeveloped areas

Factors influencing Indonesian economy

For each factor, an indicator variable equals 1 if the respondent considers the factor to play a “big” or “very big” role in influencing the Indonesian economy.

1. Laws and rules
2. Human and nature
3. World events
4. Fate

Policy support

For each policy idea, an indicator variable equals 1 if the respondent is “supportive” or “very supportive” of the policy idea.

1. Fuel and electricity price hikes accompanied by cash transfers to all, not just the poor
2. A larger government budget for the development of a mass, integrated, and environmentally friendly public transportation system
3. Greater government budget for the development of green technology and renewable energy
4. Residents displaced by infrastructure development projects benefit from the use of the new infrastructure
5. Subsidies for energy-efficient homes and buildings
6. Greater value of social assistance
7. Broader coverage of social assistance recipients, even for the middle class
8. Higher motor vehicle tax
9. Subsidize industries to switch to green technology and energy
10. The government provides financial assistance for people who have recently lost their jobs
11. Clearing forest land for agriculture or plantations, even for the purpose of producing petroleum substitute fuels (such as biofuels/bioethanol)
12. Clearing forest land for settlement development
13. Clearing forest land for infrastructure development

Personal efficacy

For each statement, an indicator variable equals 1 if the respondent answers “many” or “very many” to the item.

1. How much can we as a society do right now to create an economy that meets everyone’s needs?
 2. How much can members of the public do right now to force the government to take steps to create an economy that meets everyone’s needs?
 3. How optimistic or pessimistic do you feel right now that the public can make the government take major steps to improve the economy?
 4. How much obligation does our society have to ensure that everyone has the opportunity to succeed?
 5. How much responsibility does our society have to ensure that everyone can live a good life?
-

C Appendix C: Regression plan

Table C.1: Regression plan of the main experiment

Outcome	Fix the distribution	No victimization	Balanced Development	Equal opportunity	OLS/2SLS	Ordered Logit
Development priority						
Increase the income or economic capability of all citizens.	Y			Y	Y	
Advancing the quality of human resources.			Y	Y	Y	
Encouraging people to make decisions about their lives democratically.		Y			Y	
Erecting infrastructure and facilities for public use.	Y		Y	Y	Y	
Provide equitable, high-quality basic services (education, health, social assistance).	Y		Y	Y	Y	
Advancing underdeveloped regions.	Y		Y	Y	Y	
Enable Indonesians to achieve the important things they choose.			Y	Y	Y	
Ensure fairness for all citizens in rights, opportunities, and the law.		Y		Y	Y	
Ensuring the next generation has a better life.		Y	Y	Y	Y	
Opinions about the economy						
If people's economic conditions are good, we can reduce dependence on economic activities that damage or pollute the environment.	Y	Y		Y	Y	Y
The government may relocate people to build public facilities for the public interest (e.g., roads, reservoirs, power plants), provided that it provides replacement housing.		Y	Y		Y	Y

Table C.1: Regression plan of the main experiment

Outcome	Fix the distribution	No victimization	Balanced Development	Equal opportunity	OLS/2SLS	Ordered Logit
As citizens, we should not just accept the government's decision.		Y			Y	Y
It is natural that there are differences in wealth within society.	Y			Y	Y	Y
Laws and rules made by the government determine the chances of people succeeding.	Y			Y	Y	Y
The choices the government makes determine who is in power in our economy.	Y			Y	Y	Y
Whom should the government prioritize?						
People with capital	Y			Y	Y	Y
People in power	Y			Y	Y	Y
People with extensive connections	Y			Y	Y	Y
People who work diligently	Y			Y	Y	Y
People who are good at seizing opportunities	Y			Y	Y	Y
People who don't give in to circumstances	Y			Y	Y	Y
People who are poor or underprivileged	Y			Y	Y	Y
People living in underdeveloped areas	Y			Y	Y	Y
Factors influencing the economy						
Laws and rules	Y			Y	Y	Y
Human and nature	Y			Y	Y	Y
World events	Y			Y	Y	Y
Fate	Y			Y	Y	Y
Policy support						
Fuel and electricity price hikes accompanied by cash transfers to all, not just the poor.	Y	Y		Y	Y	Y

Table C.1: Regression plan of the main experiment

Outcome	Fix the distribution	No victimization	Balanced Development	Equal opportunity	OLS/2SLS	Ordered Logit
A larger government budget for the development of a mass, integrated, and environmentally friendly public transportation system.	Y		Y	Y	Y	Y
A greater government budget for the development of green technology and renewable energy.		Y	Y		Y	Y
Residents displaced by infrastructure development projects benefit from the resulting infrastructure.	Y	Y	Y		Y	Y
Subsidies for energy-efficient homes and buildings.	Y		Y		Y	Y
Greater value of social assistance.	Y			Y	Y	Y
Broader coverage of social assistance recipients, even for the middle class.	Y			Y	Y	Y
Higher motor vehicle tax.	Y	Y			Y	Y
Subsidize industries to switch to green technology and energy.		Y	Y		Y	Y
The government provides financial assistance for people who have recently lost their jobs.	Y			Y	Y	Y
Clearing forest land for agriculture or plantations, even for the purpose of producing petroleum substitute fuels (such as biofuels/bioethanol).	Y	Y			Y	Y
Clearing forest land for settlement development.		Y	Y		Y	Y
Clearing forest land for infrastructure development.		Y	Y		Y	Y
Personal efficacy						

Table C.1: Regression plan of the main experiment

Outcome	Fix the distribution	No victimization	Balanced Development	Equal opportunity	OLS/2SLS	Ordered Logit
How much can we as a society do right now to create an economy that meets everyone's needs?	Y			Y	Y	Y
How much can members of the public do right now to force the government to take steps to create an economy that meets everyone's needs?	Y			Y	Y	Y
How optimistic or pessimistic do you feel right now that the public can make the government take major steps to improve the economy?	Y			Y	Y	Y
How much obligation does our society have to ensure that everyone has the opportunity to succeed?	Y			Y	Y	Y
How much responsibility does our society have to ensure that everyone can live a good life?	Y			Y	Y	Y

Table C.2: Regression plan of conjoint experiment

Attributes and levels	Base level	Interactions			
		Fix the distribution	No victimization	Balanced Development	Equal opportunity
Improvement of economic conditions					
<i>ECON1</i> : The incomes of the rich rose the most, while the incomes of others rose only modestly.	Y	Y			Y
<i>ECON2</i> : The incomes of the poor or near-poor rose the most, while the incomes of the middle and upper classes rose only modestly.		Y			Y
<i>ECON3</i> : All levels of society experienced modest income increases, but no one became poorer.		Y			Y
The rights of others					
<i>RIGHTS1</i> : To build something that will benefit many people in the future, no group should be displaced or lose their livelihood, even if they receive commensurate compensation.			Y	Y	
<i>RIGHTS2</i> : To build something that will benefit many people in the future, residents should not only be compensated appropriately but also financially benefit from the project.			Y	Y	
<i>RIGHTS3</i> : To build something that will benefit many people in the future, it is natural that some residents will be forced to be evicted or lose their livelihoods, provided they receive commensurate compensation.	Y		Y	Y	
Environmental preservation					
<i>ENVI</i> : There should be no environmental destruction at all, even if it is to build something that will benefit many people in the future.	Y		Y	Y	
<i>ENV2</i> : It is acceptable to destroy the environment to some extent, as long as it is to build something that will benefit many people in the future.			Y	Y	
Citizen participation					

Table C.2: Regression plan of conjoint experiment

Attributes and levels	Base level	Interactions		
		Fix the distribution	No victimization	Balanced Development Equal opportunity
<i>PARTICIPATION1</i> : Citizens are comfortable and free to provide input, ask questions, or submit complaints to the government, so that development programs can proceed more carefully.			Y	
<i>PARTICIPATION2</i> : After voting in the election, citizens entrust the government with discretion, allowing development programs to proceed more smoothly and efficiently.	Y		Y	

D Appendix D: Additional figures

Figure D.1: Stimulus word count and comprehension

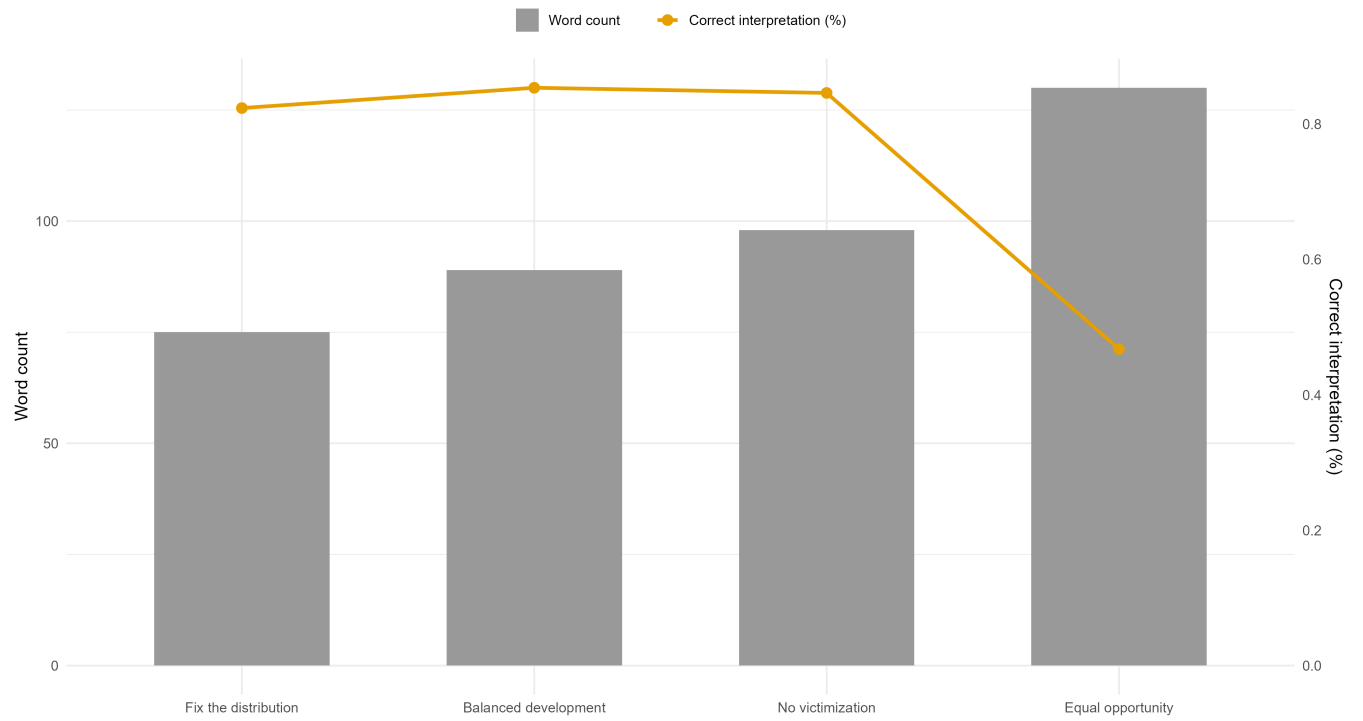


Figure D.2: Treatment effects on beliefs about whom the government should prioritize

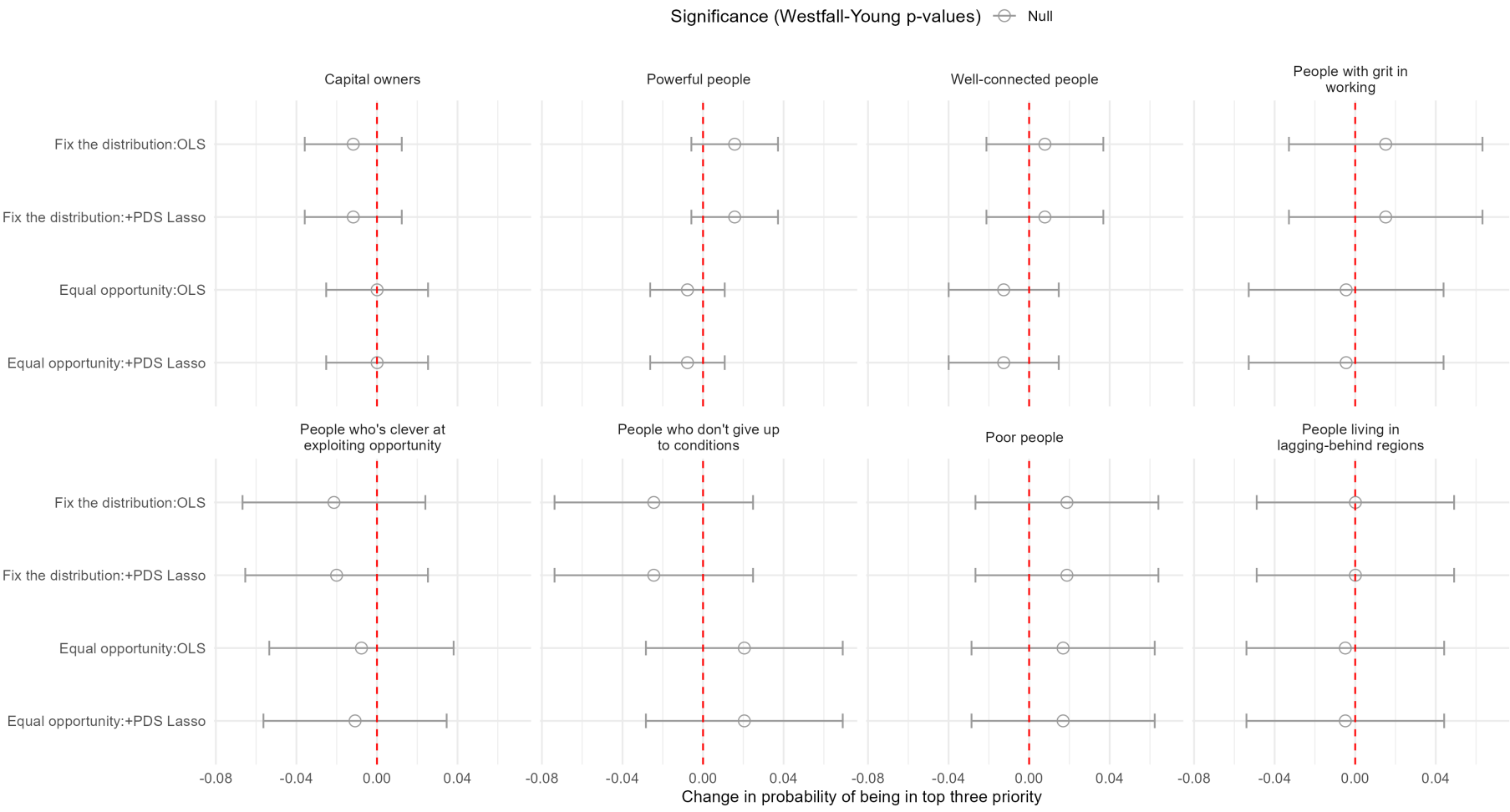


Figure D.3: Treatment effects on beliefs about factors influencing the economy

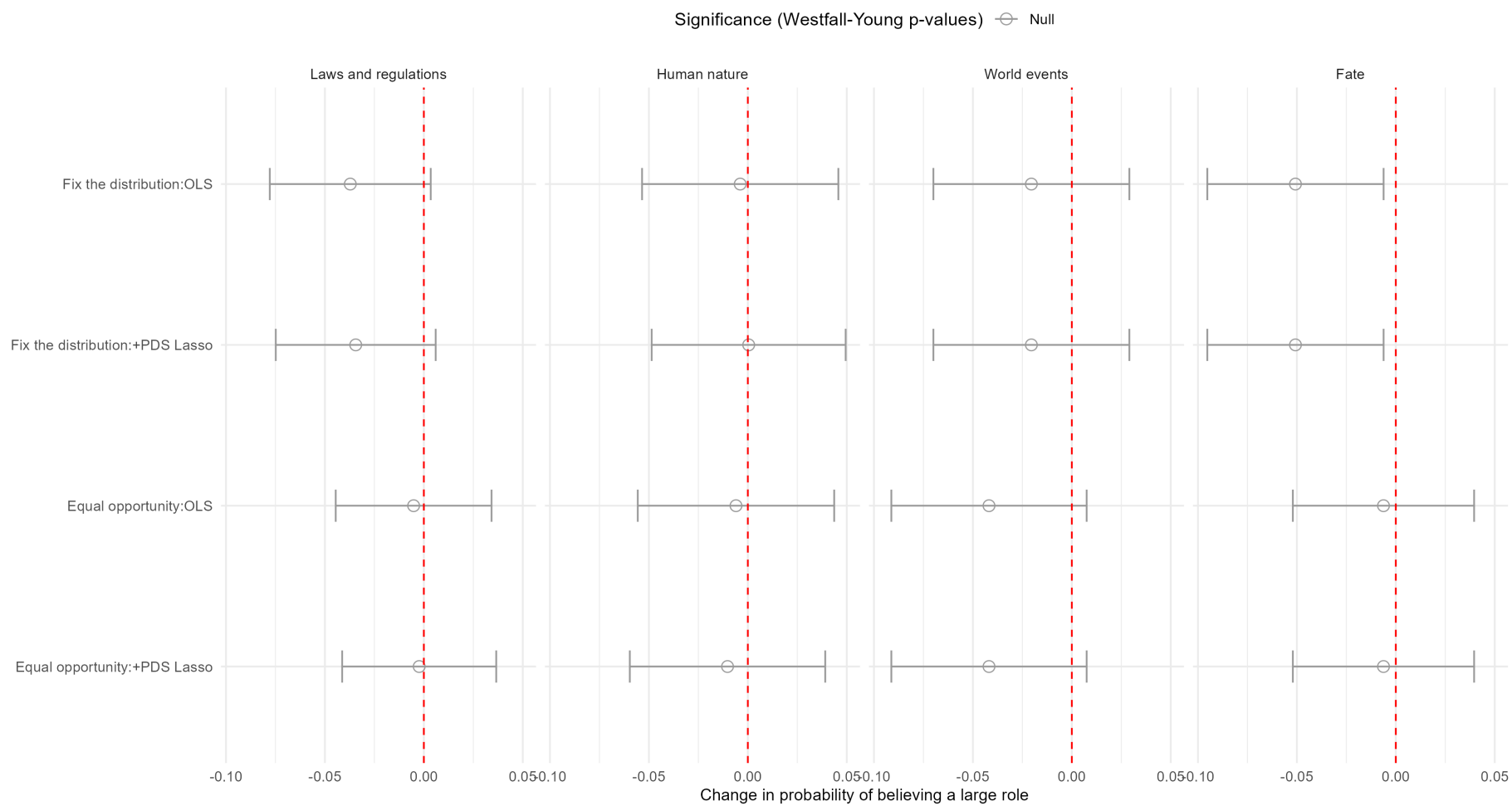


Figure D.4: Treatment effects on personal efficacy

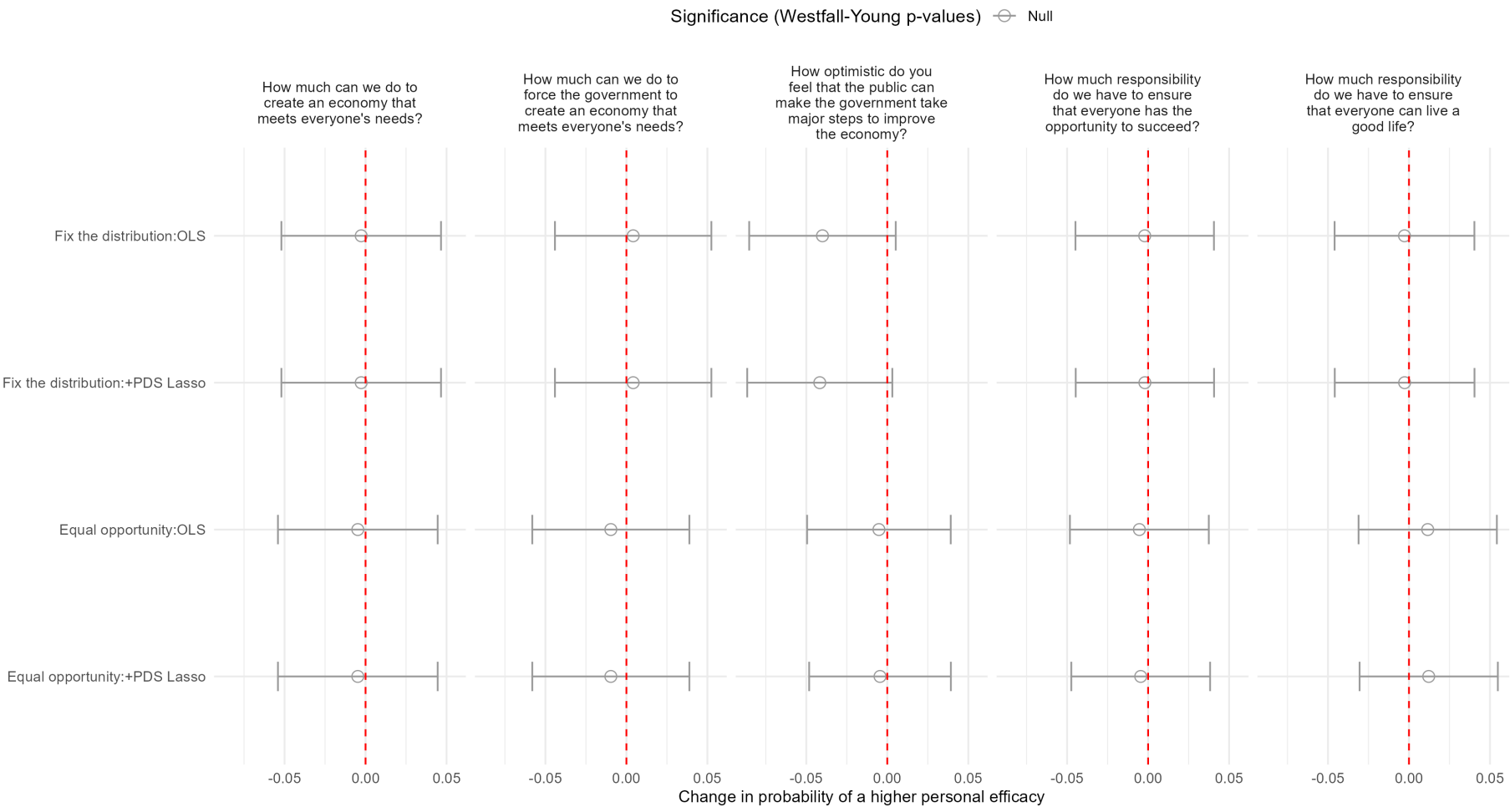


Figure D.5: Decision weight when choosing development profile

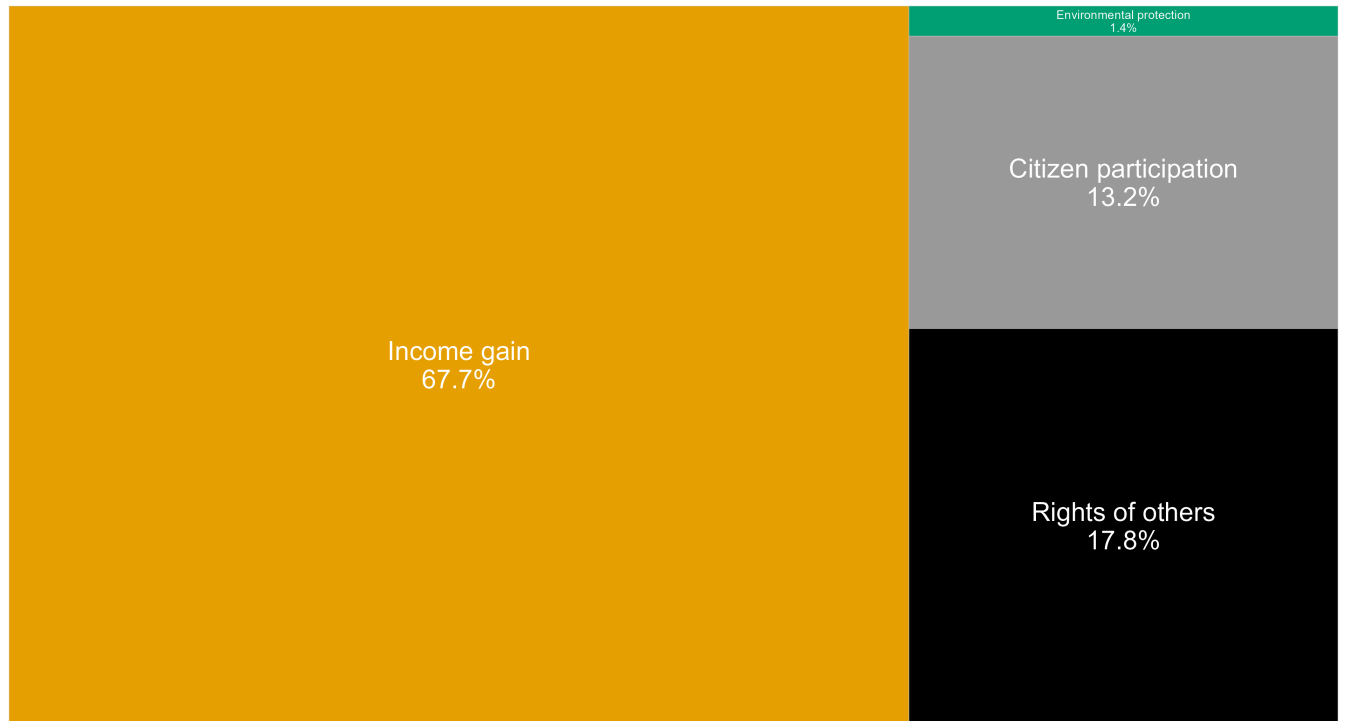


Figure D.6: Relationship between support for selected policy ideas and the most resonating paragraph from “Fix the distribution”

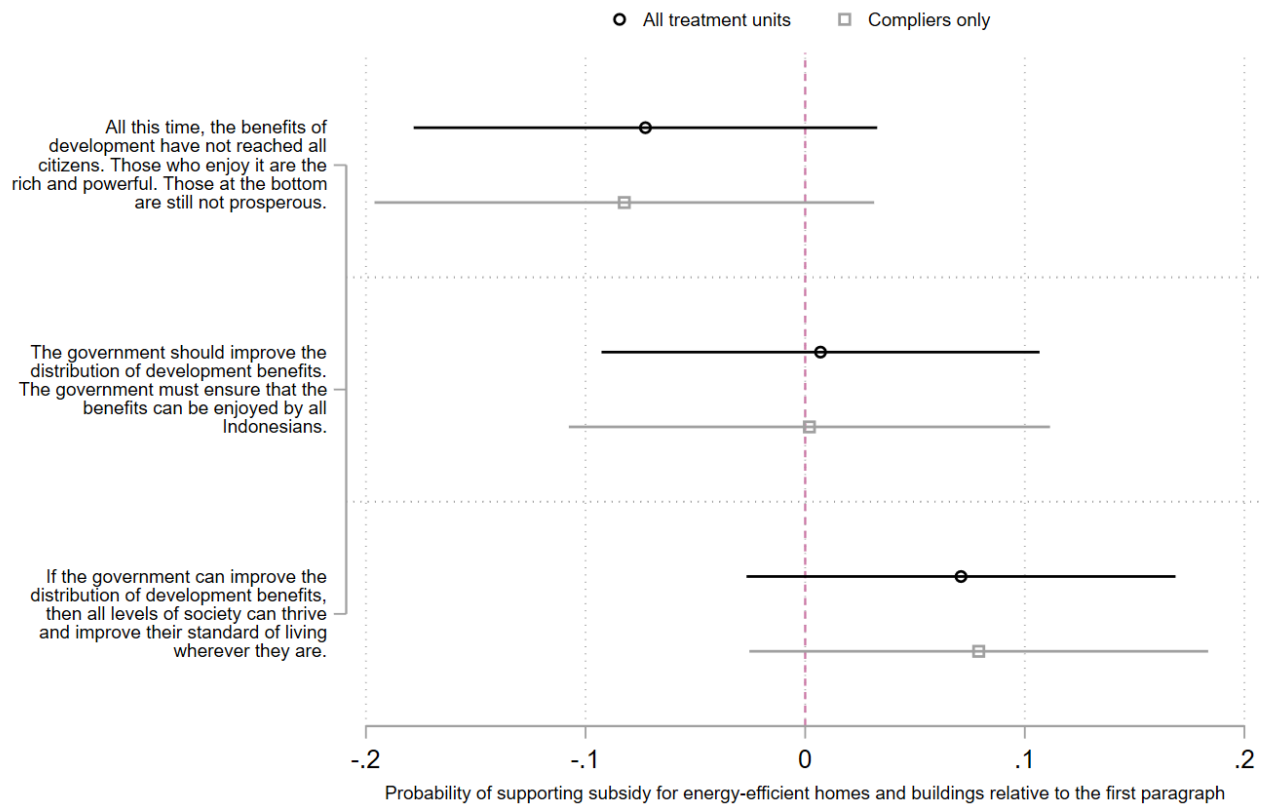


Figure D.7: Relationship between support for selected policy ideas and the most resonating paragraph from “Balanced development”

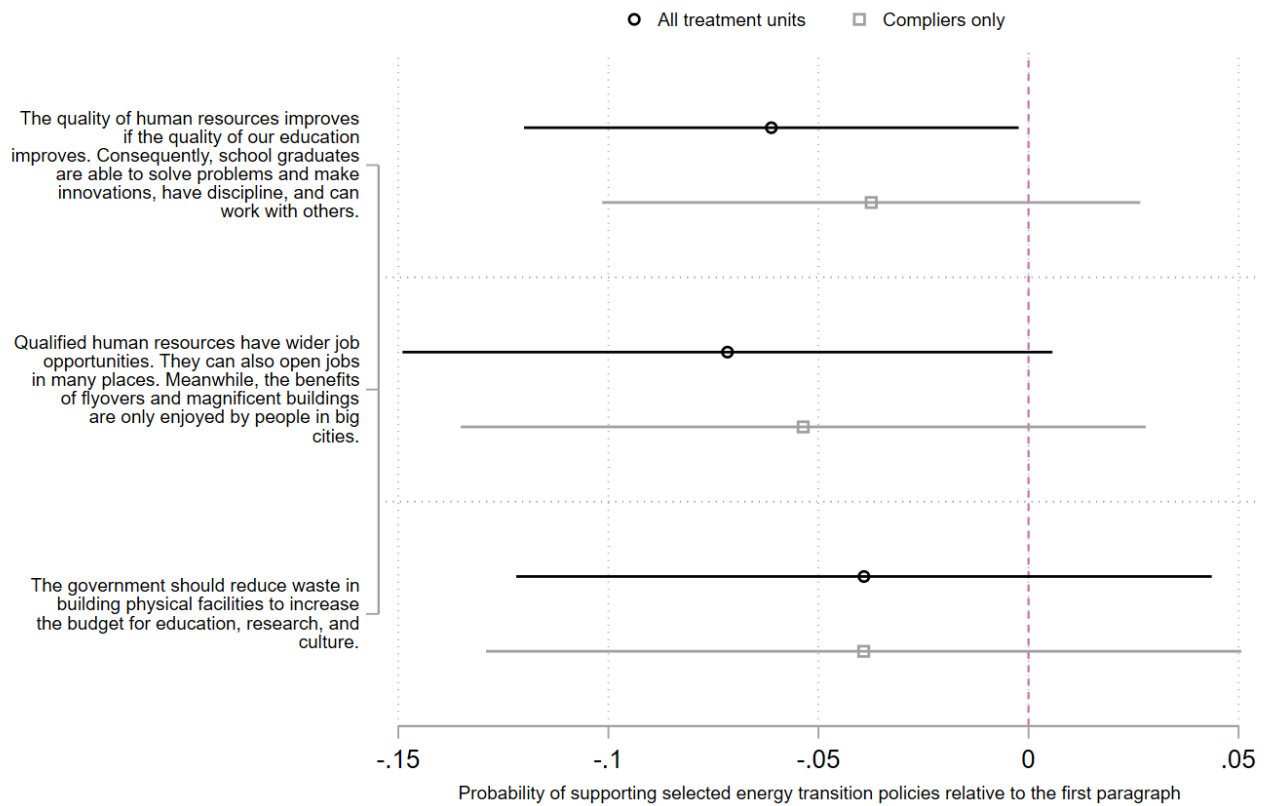


Figure D.8: Relationship between prioritization of selected development goal and most resonating paragraph from “Equal opportunity”

